

Governance effect of digital washing on corporate sustainable growth

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ABSTRACT

This study examines the governance effect of digital washing on corporate sustainable development. Although digital transformation is widely regarded as a driver of long-term value creation, some firms engage in symbolic digital narratives that exaggerate their digital capabilities without substantive operational changes, raising concerns about the implications for corporate sustainable growth. Drawing on perspectives of governance and information asymmetry, we investigate whether and how digital washing influences firms' corporate sustainable growth outcomes. Utilizing a panel of Chinese A-share-listed firms from 2014 to 2023, we find that digital washing is significantly negatively associated with corporate sustainable growth, suggesting that symbolic digital signaling undermines long-term value creation. Furthermore, this negative effect is contingent on firms' governance and information environments. Specifically, technical efficiency, knowledge value, and investor attention condition the extent to which digital washing impacts corporate sustainable growth. By distinguishing digital washing from substantive digital transformation, this study enhances the understanding of the adverse effects of digital narratives and contributes to the literature on digitalization and corporate sustainable growth. Our findings highlight the risks associated with superficial digital communication and underscore the importance of aligning digital disclosures with genuine operational capabilities to achieve corporate sustainable growth.

1. Introduction

In the context of significant global economic and social transformations, corporate sustainable development has emerged as a core issue in corporate strategy. Corporate sustainable development encompasses the comprehensive performance of enterprises across environmental, social, and economic dimensions. It is essential for maintaining long-term competitive advantages and social legitimacy (Jamil and Khan, 2024; Zhou et al., 2022). Extensive research indicates that enterprises that neglect corporate sustainable development often encounter reputational risks, policy constraints, and penalties in the capital market. Conversely, those that actively engage in corporate sustainable development tend to attract investor favor and achieve steady growth. In emerging markets, environmental risks and social responsibility issues are particularly pronounced. (Abideen and Fuling, 2024; Suchman, 1995a,b). Thus, corporate sustainable development is no longer an optional strategy but a vital requirement for the survival and growth of enterprises (Jamil and Khan, 2024).

Meanwhile, digital transformation is widely recognized as a vital driver for enterprises striving for corporate sustainable development. The implementation of digital technologies enhances resource allocation

efficiency, optimizes operational processes, and fosters green innovation and transparent information disclosure. These improvements reinforce enterprises' sustainable competitiveness (Orlikowski and Barley, 2001). From a theoretical perspective, digital transformation is regarded as a fundamental reconfiguration of organizational processes, structures, and value-creation logic, facilitated by digital technologies rather than merely the adoption of information systems (Brynjolfsson, 2014; Vial, 2021). These frameworks highlight that digital transformation reshapes managerial decision-making, coordination mechanisms, and performance evaluation criteria. This has profound implications for long-term corporate development. Existing research indicates that digitalization has significantly enhanced environmental and social performance across various industries, including textiles, manufacturing, and finance (Ji et al., 2024; Tolentino-Zondervan and DiVito, 2024). However, scholars have found that digitalization is not a panacea. To align with societal expectations regarding "digital transformation enhancing innovation capabilities," enterprises often issue digital strategy announcements and corporate social responsibility reports. These reports demonstrate their compliance with norms and alignment with external values. This strategy serves to release legitimacy signals (Suchman, 1995a,b), thereby strengthening their core competitiveness. Managers often perceive new

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technologies as a ‘panacea,’ cultivating an image of digital pioneers through successful case studies. However, extensive authoritative research across various industries consistently indicates that approximately 70% of digital transformation initiatives fail to meet their intended objectives (De la Boutetière et al., 2018; Tabrizi et al., 2019). A converging body of evidence indicates that approximately 65–70% of digital transformations fail to achieve their objectives. Harvard Business Review (HBR) reports typical failure rates around 70%, with some surveys indicating rates as high as 70–95%. Similarly, McKinsey notes that about 70% of transformations fail, while Boston Consulting Group (BCG) finds that only approximately 30% succeed. This outcome significantly increases the pressure on enterprises to fulfill their legitimacy and performance commitments. Consequently, under such pressure, many enterprises have adopted signal manipulation strategies. They selectively amplify positive signals such as investments in digital technology and R&D expenditures to gain recognition from investors and the market. This approach enhances their short-term market valuation and performance (Fioravante, 2024; Filatotchev et al., 2025; Ji et al., 2024). However, excessive reliance on or superficial implementation of digitalization efforts, without substantial improvements, may devolve into what is termed ‘digital washing.’ Such practices not only undermine the positive impact of digitalization on corporate sustainable development but also exacerbate the legitimacy risks faced by enterprises (Oludapo et al., 2024; Zhang et al., 2023). Therefore, exploring the relationship between digital washing and corporate sustainable development, as well as revealing the potential risks and regulatory mechanisms, has emerged as a significant issue of mutual concern in both academia and practice (Cao et al., 2022; Filatotchev et al., 2025; Orlikowski and Barley, 2001). Such expectations are further supported by extensive discussions on policy and governance related to responsible digital practices and transparency, as evidenced by the OECD and the European Commission. Together, these developments indicate that although digital transformation is widely advocated as a pathway to corporate sustainable development, increasing legitimacy pressures and high failure rates have fostered conditions under which symbolic digital narratives may proliferate.

Digital washing is a distorted phenomenon within the context of digital transformation. Enterprises exaggerate or fabricate the extent of their digital advancements to gain external legitimacy and market recognition, without making substantial investments in or improvements to their core capabilities (Fioravante, 2024). This behavior closely resembles traditional greenwashing. Both phenomena respond to regulatory and investor pressures through symbolic disclosures, narrative amplifications, and metric manipulations. These actions may yield short-term reputational premiums but ultimately undermine the efficiency of resource allocation and social trust in the long run (Zhang et al., 2023; Nygaard and Silkoset, 2023). Digital performance indicators are inherently more ambiguous and manipulable compared to traditional green metrics (Florian et al., 2022). Existing studies generally indicate that washing behaviors can have detrimental effects on corporate sustainable development. On one hand, they lead to resource misallocation by diverting funds that could have been used for technological upgrades or innovation toward image-building projects. On the other hand, these behaviors erode stakeholder trust and increase future financing and compliance costs for enterprises (Jamil and Khan, 2024; Zhang et al., 2023). However, from a critical perspective, the harm caused by digital washing may be even more insidious and complex. Research indicates that digital channels exhibit widespread coverage and rapid diffusion, pose challenges for governance identification, show strong dynamism and algorithmic amplification, and demonstrate high manipulability (Bak-Coleman et al., 2022; Huszár et al., 2022; McLoughlin et al., 2024; Chaudhuri and Roy, 2024). These factors may make digitalization measurement more susceptible to manipulation and subject to verification lags. This provides enterprises with greater operational leeway, which can lead to a ‘decoupling of words and actions,’ managerial disorder, and the erosion of long-term sustainable

strategies.

Despite growing attention to digital transformation, the existing literature leaves several critical gaps that motivate this study. First, most research on digitalization implicitly assumes that digital disclosures mirror substantive capability building, leaving limited conceptual and empirical separation between genuine digital transformation and symbolic digital narratives (i.e., digital washing) (Liu et al., 2023). Second, evidence on the consequences of digital washing remains scarce and is largely confined to short-term market reactions or accounting outcomes, which leaves open whether digital washing erodes firms’ long-term corporate sustainable growth trajectories (Wei et al., 2025). Third, prior work rarely examines when digital washing is most damaging—namely, how firms’ capability, governance, and information environments shape stakeholders’ ability to detect and discipline symbolic digital claims. These gaps are especially salient in emerging markets such as China, where rapid digitalization policies coexist with heterogeneous governance quality and disclosure practices. Accordingly, this study addresses two questions: (RQ1) Does digital washing undermine corporate sustainable growth? and (RQ2) Under what conditions—technical efficiency, information disclosure quality, and investor attention—is this negative effect amplified?

Although digital washing shares common institutional foundations with greenwashing—such as symbolic disclosure, legitimacy pressure, and the decoupling between words and actions—it cannot be reduced to a mere contextual extension. The distinctiveness of digital washing arises from the intrinsic features of digitalization. First, digital transformation is characterized by algorithmic opacity. Digital capabilities are embedded in complex information systems and proprietary technologies that are challenging for external stakeholders to verify. Second, digital initiatives involve a notable verification lag. The performance outcomes of digital investments often materialize only after extended implementation cycles. This allows symbolic digital narratives to persist in the short term. Third, digital narratives are highly scalable. Firms can rapidly disseminate standardized digital rhetoric across multiple disclosure channels at a low cost. These features suggest that digital washing represents a digitally enabled form of symbolic governance. It operates through mechanisms distinct from traditional greenwashing and warrants independent theoretical and empirical examination.

To address these gaps, this study makes three primary contributions. First, it focuses on digital washing and clarifies its adverse effects on corporate sustainable development. It further addresses the critical question of when these effects are most pronounced. Theoretically, the study constructs a three-dimensional analytical framework that incorporates capability, governance, and supervision. This advances the discourse on the boundary conditions of decoupling between words and actions within institutional theory and impression management (Meyer and Rowan, 1977; Velte, 2023). This development shifts the research focus from merely questioning the existence of digital washing to investigating its underlying conditions and mechanisms. Methodologically, the study introduces a reproducible measurement system and identification approach. This lays the foundation for future research to produce comparable empirical evidence. Practically, the paper provides actionable early warning indicators for regulatory authorities to identify and mitigate exaggerated or misleading digital-related disclosures. It also offers decision-making frameworks for enterprises to guide resource allocation and evaluate performance in the context of digitalization versus corporate sustainable development. Additionally, it presents criteria for investors and analysts to distinguish between symbolic displays and substantive improvements (Ji et al., 2024). In summary, this paper contributes incremental value in four dimensions: conceptual definition, mechanism explanation, quantitative identification, and policy implications. Empirically, we further document a counterintuitive governance pattern—capability, transparency, and attention can amplify (rather than buffer) the long-term costs of symbolic digital narratives.

To empirically examine these arguments, this study uses panel data

from A-share-listed companies in China from 2014 to 2023. A two-way fixed-effects model is employed to construct a three-dimensional analytical framework encompassing capability, governance, and supervision. First, we conduct a text analysis to establish and quantify a digital washing indicator system. This approach reveals the potential risks and impact mechanisms of digital washing in China from a long-term corporate strategy perspective. Second, we systematically investigate the mechanisms and boundaries of digital washing's effects on corporate sustainable development. The internal efficiency dimension underscores the necessity for enterprises to enhance their technological capabilities to ensure effective digital transformation (Weritz et al., 2025). In contrast, the external governance dimension emphasizes the roles of the market, regulation, and stakeholders. This highlights the importance of eliminating transformation bubbles through transparency of information and supervisory pressure (Grewal et al., 2021; Luo and Liu, 2024).

In this study, we define corporate sustainable development (SG) as a firm's long-term capacity to achieve stable economic growth while simultaneously maintaining environmental responsibility and creating social value. Unlike short-term performance indicators, SG emphasizes sustained value creation within the constraints of governance and strategy. Although prior studies often use related terms such as sustainability, sustainable performance, or sustainable outcomes interchangeably, we adopt corporate sustainable development as an integrative construct that encapsulates the long-term developmental consequences of corporate behavior. For clarity and consistency, the term corporate sustainable development will be used throughout this paper.

Conceptually, this definition aligns with established sustainability frameworks such as ESG, the triple bottom line (TBL), and the United Nations Sustainable Development Goals (SDGs). All of these frameworks emphasize integrating economic viability, social responsibility, and environmental stewardship. Rather than treating sustainability as a single-dimensional outcome, these frameworks conceptualize sustainable development as a long-term process constrained by governance capacity and strategic resource allocation. In this sense, the corporate sustainable development measure adopted in this study captures the economic persistence required to support environmental and social commitments over time. It thus provides an outcome-oriented proxy consistent with these broader sustainability frameworks.

2. Literature review

2.1. Digital washing undermines the corporate sustainable growth

Digital washing refers to the act of exaggerating or fabricating digital investments and achievements to create an image of advancement (Wei et al., 2025). Companies may engage in symbolic digital actions to divert public attention, aiming to maintain corporate legitimacy rather than genuinely improving corporate sustainable growth (Filatotchev et al., 2025). Legitimacy theory posits that corporate behavior is driven not only by economic objectives but also constrained by social expectations and institutional environments (Meyer and Rowan, 1977; Suchman, 1995a,b). Under external institutional and societal pressures, firms often engage in symbolic digital actions to maintain legitimacy, resulting in a “decoupling of saying and doing.” Signal theory further elucidates that companies convey signals of being “advanced,” “innovative,” and “sustainable” to the outside world through exaggerated digital disclosures and discourse strategies (Spence, 1973; Connelly et al., 2011). However, these signals are often designed for impression management and do not necessarily reflect actual digital capabilities or performance. Consequently, digital disclosures by firms become a means of “legitimacy display” rather than tools for strategic transformation. Resource misallocation theory indicates that when corporate resources are allocated to “symbolic investments” rather than “capability building,” it undermines technological efficiency and long-term value creation (Hsieh and Klenow, 2009). Digital washing distorts corporate

sustainable growth not merely through resource misallocation, but by fundamentally altering firms' incentive and governance structures. By emphasizing symbolic digital narratives over substantive transformation, digital washing reshapes internal decision-making criteria, prioritizing short-term legitimacy maintenance and external signaling over long-term capability accumulation. This shift weakens the feedback mechanisms that typically discipline inefficient strategies, delays corrective governance responses, and reduces managerial accountability for substantive performance outcomes. Consequently, strategic decisions become increasingly decoupled from long-term sustainability objectives, generating persistent governance inefficiencies that undermine corporate sustainable growth beyond mere allocation effects. Based on this, this study proposes the following hypothesis.

H1. Digital washing negatively impacts the corporate sustainable growth.

2.2. Moderating role of technical efficiency

In this study, technical efficiency is conceptualized as a substantive organizational capability that reflects a firm's ability to transform technological inputs into effective outputs. Importantly, technical efficiency is not merely an impression management or symbolic tool. However, under conditions of digital washing, high levels of technical efficiency may impair external stakeholders' ability to differentiate between substantive digital transformation and symbolic digital narratives, thereby exacerbating the governance consequences of digital washing. Technical Efficiency measures a firm's ability to achieve maximum output with given resource inputs, reflecting the core of organizational operations and resource transformation (Tone and Tsutsui, 2009). The Strategic Fit Theory posits that a firm's strategic performance is contingent upon the alignment between strategic orientation and internal capabilities (Venkatraman and Camillus, 1984). When digital strategies lack alignment with technological efficiency, digital resources may be employed for symbolic engineering rather than productivity enhancement (Ji et al., 2024). Technological efficiency should ideally facilitate the endogenous transformation of digital outcomes; however, in legitimacy-driven contexts, it may be perceived by external stakeholders as evidence of advanced digital capability, even when such capability is not substantively integrated into organizational processes. Resource misalignment theory further elucidates the intrinsic mechanisms behind this phenomenon: under governance pressure, firms may prioritize symbolic digital narratives, while high levels of technological efficiency can obscure external stakeholders' ability to assess whether these narratives are genuinely supported by the optimization of production systems and innovation processes (Benner and Tushman, 2003). Consequently, technological efficiency does not act as a buffer between digital washing and corporate sustainable growth, but may instead amplify its negative effects, creating a mechanism chain of ‘capability misinterpretation—resource misalignment—performance damage’. Fioravante (2024) emphasizes that focusing solely on “business cases” (such as efficiency and profits) while neglecting genuine social responsibilities will trigger conflicts between organizational performance and ethical outcomes, jeopardizing the long-term sustainable survival of enterprises. Based on this, this study proposes the following hypothesis.

H2a. Technical Efficiency intensifies the negative effect of digital washing on the corporate sustainable growth.

2.3. The moderating role of information disclosure quality

Information disclosure quality refers to the extent to which a company discloses information in areas related to the environment, society, and governance, reflecting its external transparency and governance standardization (Akerlof, 1978; Ioannou and Serafeim, 2015; Kuzma and Sehnem, 2023). The theory of information disclosure quality that

transparency can enhance the accountability of investors and the public, thereby reducing information asymmetry (Healy and Palepu, 2001). Prior research indicates that a high level of disclosure helps bridge the information gap between companies and investors, improving the market's capacity to recognize corporate behaviors. High-quality information disclosure can reduce capital costs (Chen et al., 2024), improve ESG transparency (He and Ismail, 2024; Yuan et al., 2025), and enhance external trust (García-Sánchez et al., 2019; Y. Liu et al., 2023). However, under external pressures and scrutiny from capital markets, companies often adopt “symbolic” governance commitments to garner positive reactions rather than effecting genuine change (Westphal and Zajac, 1998; Papa et al., 2024, Neokleous and Elmarzouky, 2025). According to agency theory (Jensen and Meckling, 1976), in the context of digital washing, management aligns with market expectations by formalizing information disclosure and enhancing the ‘digital narrative.’ This approach will draw external attention to the company's digital commitments. High-disclosure corporate behaviors are more easily cross-verified and tracked, making deceptive digital strategies more likely to be exposed (Grewal et al., 2021). Therefore, when digital washing occurs, such exposure intensifies the trust crisis and may trigger strategic deviations. Based on this, this study proposes the following hypothesis.

H2b. The Enhancement of information disclosure quality exacerbates the negative impact of digital washing on the corporate sustainable growth.

2.4. Moderating role of investor attention

We conceptualize investor attention as a dynamic governance mechanism rather than merely an impression management tool. In the initial stages, heightened attention reshapes firms' disclosure incentives by increasing their sensitivity to market evaluations, which may temporarily amplify the effectiveness of symbolic digital narratives. Over time, as attention persists and information becomes more transparent, sustained scrutiny enhances stakeholders' ability to identify inconsistencies between digital narratives and substantive actions, thereby strengthening disciplinary and reputational consequences. Investor Attention refers to the breadth and depth of the perception of capital market participants regarding the information of a particular enterprise (Da et al., 2011). Concentrated attention can enhance an individual's information processing and recognition capabilities (Kahneman, 1973), however, conversely, every move of an enterprise can be magnified and scrutinized. Investor attention significantly shapes firms' informational environments by influencing the observation, interpretation, and evaluation of corporate behaviors by capital market participants (Barber and Odean, 2008). The Attention-Based View posits that organizational decision-making is contingent upon the allocation of managerial attention (Ocasio, 1997). When investor attention increases, management's focus is often dominated by short-term market evaluations and external reputations, leading to resource misallocation (Ju et al., 2024). Furthermore, the mechanism of media and analyst pressure exacerbates this trend: heightened visibility and public scrutiny compel firms to solidify investor trust through “showable digital outcomes” (Healy and Palepu, 2001). However, when a highly attentive group of investors identifies and exposes a company's digital washing behaviors, it can trigger punitive reactions such as stock sell-offs. Under heightened investor attention, firms encounter a more pronounced evaluative environment, which may alter disclosure incentives and increase reliance on symbolic digital narratives in the short term (Long et al., 2025). As attention persists and information accumulates, investor scrutiny may intensify, thereby increasing the likelihood that inconsistencies between digital narratives and substantive actions are identified and addressed. In the long term, this can exacerbate the negative impacts on

a company's corporate sustainable development (Abideen and Fuling, 2024; Fioravante, 2024). Consequently, this study proposes the following hypothesis.

H2c. Investor Attention exacerbates the negative effect of digital washing on the corporate sustainable growth.

Taken together, the above literature reveals important gaps that motivate the present study and lead directly to the research questions examined below. Specifically, this study addresses two research questions: (RQ1) Does digital washing undermine corporate sustainable growth? and (RQ2) Under what conditions—technical efficiency, information disclosure quality, and investor attention—is this effect amplified?

3. Data and model description

This study adopts a quantitative panel-data research design to examine the long-term governance effects of digital washing on corporate sustainable growth. The empirical strategy is designed to ensure both measurement validity and causal identification. First, Chinese A-share listed firms are selected as the research context due to their intensive digitalization policies, relatively high disclosure incentives, and substantial heterogeneity in governance and information environments. Second, the data collection and variable construction procedures are grounded in established empirical practices, combining text-based measures, efficiency estimation, and market-based indicators to capture both symbolic digital behavior and substantive firm characteristics. Third, the baseline analysis employs a two-way fixed-effects model to control for unobserved firm heterogeneity and common time shocks, while interaction specifications are used to examine conditional effects. Finally, a series of robustness checks—including alternative variable constructions, propensity score matching, instrumental variable estimation, and dynamic specifications—are conducted to mitigate potential endogeneity concerns and to strengthen the credibility of the empirical findings.

3.1. Data sources and sample selection

This study investigates Chinese A-share listed firms over the period 2014–2023. Core financial indicators were extracted from the China Stock Market & Accounting Research (CSMAR) database, supplemented with textual analysis of annual reports and cross-validation with patent data. Referring to Wei et al. (2025), the sample selection adopts the following processing criteria: (1) excluding the financial industry and ST/*ST companies; (2) removing observations with missing values; and (3) mean imputation for control variables and winsorizing continuous variables at the 1st and 99th percentiles. A total of 4566 companies, with 24,483 observations, was obtained. Finally, we calculated the digital washing measure and excluded observations with residuals less than or equal to 0, ultimately obtaining 8237 observations from 2377 companies.

3.2. Variable definitions and descriptions

3.2.1. Dependent variable

Corporate sustainable growth indicator served as the explanatory variable. The corporate sustainable growth can be measured in various ways. Higgins and Van Horne's sustainable growth model is a widely accepted approach. Although this model is simple and easy to operate, it overlooks additional stock issuance factors that are inconsistent with the actual environment. Therefore, referring to the research findings of the Van Horn corporate sustainable growth model was adopted to measure corporate sustainable growth (Xu et al., 2021). The larger the value calculated by the formula, the stronger the corporate sustainable growth

capability of the enterprise.

overall length of the annual report (see Table 2).

Based on Richardson's research, we used a formula to measure the

Sustainable development of the enterprise = *Net sales interest rate* * *Earnings retention rate*

$$\frac{(1 + \text{Equity ratio})}{\left(\frac{1}{\text{Total}} \text{asset turnover rate} - \text{Net sales interest rate} * \text{Earnings retention rate} * (1 + \text{Equity ratio})\right)}$$

3.2.2. Independent variable

The Digital Washing Index is constructed using the residual method. The degree of digital transformation of a firm in year t is regressed against the benchmark level of the same industry for that year, and the residual is extracted as the measure of digital washing. This residual reflects the deviation of the firm's digital narrative from a reasonable benchmark; a larger residual indicates a higher likelihood of misrepresentation or exaggeration of the firm's level of digital transformation (Wei et al., 2025). To construct the digital transformation indicator (Dig), we initially extracted terms related to digital technology, big data technology, artificial intelligence technology, cloud computing technology, and digital technology applications from government policy documents through contextual analysis. These terms were utilized to create a dictionary of corporate digital transformation keywords. Following the establishment of this preliminary dictionary, we implemented Python-based word segmentation and conducted manual screening to eliminate irrelevant terms, retaining only those that appeared at least ten times. The filtered terms were subsequently matched with digital transformation keywords from the CSMAR database, and overlapping terms were preserved to form the final vocabulary, as presented in Table 1. Using this dictionary, we conducted a rule-based text analysis on the full texts of firms' annual reports and calculated the total frequency of digital transformation terms. Subsequently, the enterprise-level digital transformation index (Dig) was obtained by dividing the total frequency of digital-related terms by the

digital washing variable. Considering the clustering effect of digital transformation, we incorporated firm size (Size), leverage ratio (Lev), age (Age), institutional ownership ratio (Inst), proportion of independent directors (Indp), ownership concentration (Top1), and price-to-book ratio (PB) as control variables (Richardson, 2006).Based on Dig, the degree of digital transformation MDT of other enterprises in the same industry and the same year is derived and substituted into the formula to calculate the residual ε_{it} . The residual ε_{it} represents the abnormal degree of digital disclosure, namely digital washing. To mitigate issues of serial correlation and within-group correlation, we controlled for firm-level clustering effects and incorporated fixed effects for both year and industry. When digital washing is greater than 0, it indicates the presence of "digital washing," and the higher the digital washing, the broader the scope of digital washing. When digital washing is greater than 0, it indicates the presence of "digital washing," and the higher the digital washing, the broader the scope of digital washing. To address issues of serial correlation and within-group correlation, standard errors are clustered at the firm level, and both industry and year fixed effects are incorporated in all regressions.

$$Dig_{it} = \beta_0 + \beta_1 MDT_{it} + \sum_{it} Control + \sum_{it} id + \sum_{it} Year + \varepsilon_{it}$$

Here, i represents the enterprise and t represents the year. Dig_{it} represents the degree of digital transformation of enterprise i in year t; MDT_{it} represents the degree of digital transformation of other enterprises in the same industry and the same year; the residual ε_{it} represents the abnormal degree of digital disclosure.

Table 1
Digital transformation keywords dictionary.

Dimension	Classifier term
Big Data Technology	Big Data, Mixed Reality, Data Visualization, Data Mining, Text Mining, Virtual Reality, Heterogeneous Data, Augmented Reality, Credit Investigation.
Blockchain Technology	Bitcoin, Differential Privacy Technology, Distributed Computing, Consensus Mechanism, Consortium Blockchain, Decentralization, Digital Currency, Smart Contract.
Artificial Intelligence Technology	Machine Learning, Artificial Intelligence, Facial Recognition, Business Intelligence, Identity Verification, Deep Learning, Biometric Technology, Image Understanding, Semantic Search, Speech Recognition, Intelligent Robotics, Intelligent Data Analysis, Autonomous Driving, Natural Language Processing.
Digital Technology Application	B2B, B2C, C2B, C2C, Fintech, NFC payment, O2O, third-party payment, e-commerce, industrial internet, internet finance, internet healthcare, financial technology, open banking, quantitative finance, digital finance, digital marketing, Nets Union, unmanned retail, mobile internet, mobile internet, mobile payment, smart agriculture, wearable technology, smart grid, smart environmental protection, smart home, smart transportation, intelligent customer service, smart energy, robo-advisor, smart cultural tourism, smart healthcare, intelligent marketing.
Cloud Computing Technology	EB-level storage, multi-party secure computation, brain-inspired computing, stream computing, green computing, in-memory computing, cognitive computing, converged architecture, graph computing, Internet of Things, cyber-physical systems, hundred-million-level concurrency, cloud computing.

3.2.3. Independent variable

3.2.3.1. Technical Efficiency. We employed the Stochastic Frontier Analysis (SFA) to measure the technical efficiency of enterprises. SFA can reveal the stochastic disturbances faced by enterprises in the production process and the deviation in technical efficiency by modeling the production process. Compared to non-parametric Data Envelopment Analysis (DEA), the primary advantage of Stochastic Frontier Analysis (SFA) lies in its capacity to differentiate between inefficiency and statistical noise or measurement errors. This capability helps to prevent the misclassification of random disturbances as efficiency losses. In situations characterized by uncertainty or measurement errors in inputs and outputs, SFA demonstrates greater robustness than DEA (Lamb and Tee, 2024; F. Liu et al., 2023). Furthermore, SFA facilitates the inclusion of control variables and environmental factors within a parametric framework and extends to panel settings, which is essential for addressing unobserved heterogeneity and dynamic effects relevant to this study's focus on measuring technical efficiency across firms and over time (Parmeter and Kumbhakar, 2025; Strange et al., 2021). Considering these factors, we employed Stochastic Frontier Analysis (SFA) to assess firms' technical efficiency. In line with Wang and Ho (2010), we specify a panel stochastic production frontier model that incorporates time-varying inefficiency. Firm-level output is represented by operating revenue, while capital and labor inputs are proxied by fixed assets and the number of employees, respectively, consistent with existing literature on firm-level efficiency in emerging markets. We adopt a

Table 2
Variable definitions.

Variable	Abbreviation	Description
Dependent variable		corporate sustainable growth
Corporate sustainable growth	SG	Adopted the Van Horne corporate sustainable growth Model to measure
Independent variable		
Digital Washing	DW	Employed text mining and machine learning techniques to obtain the Digital Transformation Index (dig), calculated the degree of digital transformation (DMT) of other enterprises in the same industry and the same year, incorporated it into the regression to compute the residuals, and selected data greater than 0 to represent digital whitewashing.
Moderating variable		
Technological Efficiency	TE	Used Stochastic Frontier Analysis (SFA) to measure the technical efficiency of enterprises, as shown in Table 1
Information Disclosure Quality	KV	Constructed quantitative indicators of information disclosure quality using textual analysis methods
Investor Attention	IA	Constructed an attention index by integrating the search volume data of listed companies' stock codes, abbreviations, and full names, and took the natural logarithm
Control variable		
Asset Size	Size	Used the natural logarithm of total assets to measure and control the resource differences arising from scale effects
Ind Director Ratio	Indp	Indp = the number of independent directors/the total number of board members, reflecting the independence of the corporate governance structure and the capacity for external oversight
Asset Liability Ratio	Lev	Total Liabilities/Total Assets, reflecting the impact of capital structure on operational decisions
Asset Size	Age	The natural logarithm of the number of years from the establishment year to the observation year
Equity Concentration	Top1	The combined shareholding ratio of the top ten shareholders reflects the characteristics of the governance structure
Investor's Shareholding Ratio	Inst	The number of institutional shareholdings/total share capital, reflecting external oversight mechanisms
Price-to-Book Ratio	PB	PB = company market value/net assets, reflecting the market's evaluation level of a company's net assets.
Enterprise Growth Potential	Growth	Growth = (Total assets at the end of the current period - Total assets at the end of the previous period)/Total assets at the end of the previous period, reflecting the rate of change in the scale of a company's assets.
Corporate profitability	ROA	ROA = Net Profit/Total Assets, reflecting the short-term profitability of the enterprise.
Proportion of Fixed Assets	PPE	PPE = Fixed assets/Total assets, reflecting the stability of the enterprise's assets and the long-term nature of capital investment.

Cobb–Douglas functional form due to its parsimonious structure and its widespread application in panel SFA studies, which facilitates stable estimation under large sample conditions.

In the stochastic frontier specification, the composite error term comprises two components: a symmetric random noise term, which is assumed to follow a normal distribution $N(0, \sigma_v^2)$, capturing measurement errors and exogenous shocks, and a non-negative inefficiency term u_{it} , which is assumed to follow a truncated normal distribution, reflecting firm-specific technical inefficiencies. The inefficiency component is permitted to vary over time, allowing us to capture dynamic changes in firms' operational efficiency. Technical efficiency is then calculated as the exponential transformation of the inefficiency

term, with values ranging from 0 to 1, where higher values indicate greater efficiency.

Technical efficiency formula.

Variable type	Formula	Description
Forward Production Function (FPF)	$\Delta y_{it} = \Delta x_{it}^T \beta + \Delta v_{it} - \Delta u_{it}$	Structuring output and input relationships
Dynamic inefficiency term	$u_{it} = u_i \cdot \exp(\eta(t - T))$	Capturing time-varying characteristics of efficiency
Technical efficiency	$TE_{it} = \exp(-u_{it})$	Measuring Business Efficiency

$TE_{it} \in (0,1]$ represents the technical efficiency of firm i in period t . The closer the value is to one, the greater the efficiency; $u_{it} > 0$ denotes the technical inefficiency term, reflecting the extent to which a firm's actual output deviates from the frontier output. $\exp(-u_{it})$ denotes an exponential function that transforms the inefficiency term u_{it} into the efficiency value TE_{it} .

3.2.3.2. Information disclosure quality. Referencing the study by (Kim and Verrecchia, 2001), we employed text analysis to construct a quantitative measure of information disclosure quality, using the “Management Discussion and Analysis” (MD&A) section of annual reports as the source text. We operationalized disclosure quality along five dimensions: quantitative (Qn), forward-looking (Fo), strategic (St), comparable (Cm), and complete (Cmplt). Specifically, Qn quantifies the extent of verifiable numerical disclosure by counting numerical expressions (e.g., percentages, investment amounts, targets, and timelines) within digital-related sentences and normalizing this count by the total number of such sentences. Fo captures a forward-looking orientation by identifying prospective cues (e.g., “will,” “plan,” “expected,” “by [year]”) and normalizing their frequency. St assesses strategic specificity by detecting references to concrete digital initiatives and KPI-related terms (e.g., “strategy,” “roadmap,” “milestone,” “KPI/indicator,” “implementation,” “deployment”) and normalizing these references accordingly. Cm evaluates comparability by identifying benchmarking and time-consistency cues (e.g., “year-on-year,” “compared with,” “industry average,” “ranking,” “consistent standards”) that facilitate cross-firm or intertemporal comparisons. Cmplt measures completeness by assessing the breadth of coverage across key digital topics (e.g., AI, big data, cloud computing, blockchain, digital applications) and determining whether disclosures encompass both actions and outcomes/targets, with higher scores assigned to broader and more balanced coverage. Each dimension score is rescaled to a 0–1 range and standardized to ensure comparability across firm-years. The overall disclosure quality index is then computed using a weighted-average approach.

$$KV = \sum_{i=1}^n \omega_i \times D_i$$

D_i represents the score of the i -th dimension, ω_i denotes the weight of that dimension, and $\omega_i = 1$. Based on empirical literature, this study set equal weights for all dimensions, i.e., $\omega_i = 0.2$. Formally, $KV_{it} = 0.2 \times Qn_{it} + 0.2 \times Fo_{it} + 0.2 \times St_{it} + 0.2 \times Cm_{it} + 0.2 \times Cmplt_{it}$. The final information disclosure quality value ranges between 0 and 1, with higher values indicating more comprehensive and higher-quality disclosures.

3.2.3.3. Investor attention. Consistent with (Da et al., 2011), we proxy investor attention using an Internet search index. The attention metric is constructed by integrating the search volume data of the stock codes, abbreviations, and full names of listed companies. Specifically, based on the “Internet Search Index Database” of the China Research Data Service Platform, the daily search volume data for stock codes, abbreviations, and full names of the sample companies from 2014 to 2023 were retrieved. The daily search volumes for the full names, abbreviations, and stock codes of individual companies were aggregated on an annual

basis to calculate the total search volume. To mitigate the effect of outliers, the natural logarithm of the total search volume was used to derive the final metric. After the log transformation, the values were significantly smaller than those of the initial data, which may have caused the coefficients to be insignificant. Referring to (Yezege, 2023; Kuo and Huang, 2022) the attention coefficient estimates in the Conclusion section were multiplied by 100 for ease of interpretation.

3.2.4. Control variables

This study selected ten control variables to account for the impact of firm heterogeneity on the research findings and to control for policy resources and agency problems arising from ownership differences (Yu and Zhang, 2025). By controlling for these core elements that may influence corporate behavior, the robustness of the model estimation can be effectively enhanced, ensuring the reliability of the research conclusions.

3.2.5. Data description

To ensure the robustness of the regression estimates, this paper standardized, log-transformed, and winsorized the primary variables. First, a natural logarithm transformation was applied to continuous variables to mitigate skewed distributions and enhance comparability. Second, to reduce the influence of extreme values on the results, all continuous variables were winsorized at the 1st and 99th percentiles, thereby eliminating the excessive impact of outliers. Third, variables were mean-centered to clarify the interpretation of interaction terms and to reduce the risk of potential multicollinearity. Furthermore, in constructing the digital washing indicators, the residuals generated from the industry and year benchmark regressions were screened, excluding those abnormal residual values that exceeded the mean ± 3 standard deviations, in order to avoid estimation bias caused by a small number of extreme firm data. The aforementioned processing methods are widely adopted in the existing literature and are deemed to enhance the interpretability and robustness of the results.

3.3. Model design

3.3.1. Benchmark regression model

Based on the results of the Hausman test, this paper employs a two-way fixed effects model that incorporates both firm and year effects to control for unobservable time-invariant firm characteristics and macroeconomic annual shocks. Standard errors are clustered at both the firm and industry levels to mitigate issues related to serial correlation and within-group correlation. This study proposed a benchmark regression model to examine the negative impact of digital washing on corporate sustainable growth.

$$SG_{it} = \beta_0 + \beta_1 DW_{it} + \sum Control_{it} + id_t + year_i + \varepsilon_{it} \quad (1)$$

In Model1, i denotes the company, and t denotes the year. SG_{it} is the dependent variable of the model, used to capture firms' long-term corporate sustainable growth, with DW_{it} being the primary explanatory variable for digital washing. β is the regression coefficient; if β_1 is significantly less than 0, it supports the proposed H1 hypothesis. Control represents the control variables. id_t and $year_i$ are the fixed effects of the model, accounting for potential impacts from time and individual companies, and ε_{it} is the random disturbance term. Prior to model estimation, all key variables were subjected to logarithmic transformation and winsorization. Additionally, samples with residuals exceeding ± 3 standard deviations were excluded from the analysis.

3.3.2. Adjustment regression model

This study constructed three moderation effect models to analyze whether the moderating variables of technical efficiency, quality of information disclosure, and investor attention moderate the impact of corporate digital washing on corporate sustainable growth.

$$SG_{it} = \beta_0 + \beta_1 DW_{it} + \beta_2 TE_{it} + \beta_3 DW \times TE_{it} + \sum Control_{it} + id_t + year_i + \varepsilon_{it} \quad (2)$$

$$SG_{it} = \beta_0 + \beta_1 DW_{it} + \beta_2 KV_{it} + \beta_3 DW \times KV_{it} + \sum Control_{it} + id_t + year_i + \varepsilon_{it} \quad (3)$$

$$SG_{it} = \beta_0 + \beta_1 DW_{it} + \beta_2 Attention_{it} + \beta_3 DW \times Attention_{it} + \sum Control_{it} + id_t + year_i + \varepsilon_{it} \quad (4)$$

In Model 2, 3, and 4, i and t denote firm and year, respectively. SG_{it} represents the short-term financial performance of the firm, DW_{it} signifies the level of digitalization of the firm, and Control is the control variable. id_t is the firm-specific fixed effect and Year is the year fixed effect. ε_{it} is the random disturbance term. $DW \times TE_{it}$, $DW \times KV_{it}$, $DW \times IA_{it}$ are the interaction terms of the moderating variables, technical efficiency, information disclosure quality, and investor attention, respectively. Where technical efficiency/information disclosure quality/investor attention is the moderator. The moderation is formally tested by the interaction coefficient β_3 via $H_0: \beta_3 = 0$. Because the marginal effect $\frac{\partial SG}{\partial DW} = \beta_1 + \beta_3 M$, we also report simple slopes at $M = \{-1\sigma, 0, +1\sigma\}$ and provide plots for interpretation.

4. Empirical results

4.1. Descriptive statistics

Table 3 presents the statistics derived from the processed data utilized for regression analysis. To ensure consistency with the regression methodology, all continuous variables, except for Inst, Top1, and PB, have been winsorized at the 1st and 99th percentiles, log-transformed, and de-meaned. Consequently, it is anticipated that the means of variables such as SG, DW, technical efficiency, information disclosure quality, and investor attention will be approximately zero. The presence of negative values merely indicates a position 'below the sample mean' following de-meaning and does not signify negative values in an economic context. Notably, some variables (Age, Indp, Top1) demonstrate high standard deviations, indicating considerable heterogeneity among firms regarding establishment years, governance structures, and ownership concentration. This variability reflects the actual characteristics of Chinese listed companies but may influence the stability of regression estimates. To mitigate this issue, the paper applies winsorization and de-meaning to the relevant variables in the main regression specification and utilizes Z-score standardization for alternative

Table 3
Descriptive Statistics derived from the processed data.

Variables	Mean	Sd	Min	Max
SG	0.000	0.062	-0.086	0.271
DW	0.000	0.000	0.000	0.001
TE	0.000	0.136	-0.157	0.768
KV	0.000	0.195	-0.363	0.618
Attention	-0.000	0.875	-2.691	2.229
Inst	0.018	0.255	-0.424	0.484
Indp	0.040	5.365	-4.598	19.212
Top1	0.014	0.156	-0.255	0.411
Size	-0.048	1.147	-2.268	3.882
Lev	-0.005	0.190	-0.347	0.456
Age	-0.000	7.565	-9.658	18.342
ROA	0.000	0.057	-0.240	0.161
PB	0.148	3.032	-3.229	13.206
PPE	0.000	0.134	-0.168	0.461
Growth	-0.000	0.221	-0.385	1.132

estimations in robustness checks (Age_z, Indp_z, Top1_z). The results indicate that the core conclusions remain consistent across various measurement methods, suggesting that the findings are robust. The distribution of variables reveals that the mean of digital washing is nearly zero with an almost negligible standard deviation, indicating that most enterprises do not exhibit significant misrepresentation in their digital transformation. However, the difference between the minimum and maximum values still suggests a certain degree of heterogeneity among enterprises. This limited yet crucial dispersion implies that a small number of enterprises exhibit a more pronounced tendency towards digital fabrication in information disclosure or strategic positioning, which may significantly influence the judgments of capital markets and external stakeholders. In comparison, while the average level of corporate sustainable growth among enterprises is generally lower, it exhibits a degree of variability across different enterprises, as indicated by the standard deviation. This reflects the differentiated performance of enterprises in addressing long-term development and digitalization pressures, and also provides the necessary statistical basis for verifying the inhibitory effect of digital washing on corporate sustainable growth.

4.2. Correlation analysis

To assess the interdependence between variables and potential multicollinearity, we conducted correlation analysis and variance inflation factor (VIF) tests. Table 4 indicates a negative correlation between digital washing and corporate sustainable growth ($r = -0.024$, $p < 0.05$), which preliminarily supports hypothesis H1. The VIF values for all variables were below 3, with an average VIF of 1.36, which was significantly lower than the critical threshold of 10. This suggests that there are no severe multicollinearity issues in the model.

4.3. Empirical analysis results

4.3.1. Baseline regression results

This study employed a two-way fixed-effects model to measure the negative impact of digital washing on corporate sustainable growth, effectively controlling for unobservable individual heterogeneity at the firm level and common external shocks over time. Compared with other models, this method can more precisely identify the net effect between digital washing and corporate sustainable growth, reduce estimation bias caused by omitted variables, and ensure the robustness of the research findings. As shown in Table 5 Model (1) Baseline, the regression coefficient of digital washing is significantly negative. The results are significant at the 5% significance level ($\beta = -7.361$, $p < 0.05$). Although the simple correlation between DW and SG is modest, this is expected given the residual-based construction of DW and the presence of substantial firm- and time-level heterogeneity. This indicates a significant correlation between a higher level of digital washing and a lower level of corporate sustainable growth. The statistical results are robust. From a theoretical standpoint, this result was consistent with the findings of Wei et al. (2025). Digital washing significantly reduces accounting conservatism. Li et al. (2024) indicate that the green marketing behavior of enterprises can enhance stock returns in the short term but will reduce the yield rate in the long run. Luo and Liu (2024) demonstrate that the genuine application of digital technology can significantly enhance the environmental and economic performance of enterprises. This promotes corporate sustainable growth. Evidently, when enterprises engage in superficial digital promotion without substantive innovation, their capacity for corporate sustainable growth is undermined. This finding reflects the severe consequences of resource misallocation and reputational damage. Digital washing, as a superficial endeavor, not only diverts resources that could otherwise support innovation but also erodes investor trust. This compounds the inhibition of firms' long-term growth potential.

Although the distribution of SG is relatively concentrated, this

Table 4
Correlation and VIF results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
DW	1.000														
SG	-0.024**	1.000													
KV	-0.012	0.211***	1.000												
TE	-0.044**	-0.029***	0.089***	1.000											
Attention	0.023	0.060***	0.179***	0.215***	1.000										
Inst	-0.185***	0.045***	0.180***	0.225***	0.088**	1.000									
Indp	0.024**	0.022*	0.030***	0.033***	0.013	-0.091***	1.000								
PB	0.113***	0.300***	0.166***	-0.055***	0.129***	-0.052***	0.031***	1.000							
Lev	-0.084**	-0.311***	-0.039***	0.259***	0.154***	0.170***	-0.032***	-0.107***	1.000						
Size	-0.080***	-0.050***	0.253***	0.399***	0.327***	0.430***	-0.001	-0.259***	0.409***	1.000					
Top1	-0.183***	0.024*	0.026**	0.159***	-0.006	0.528***	0.028**	-0.055***	0.044***	0.174***	1.000				
Age	-0.115***	-0.099***	0.022	0.062**	0.237***	0.208***	-0.031***	-0.194***	0.264***	0.378***	-0.010	1.000			
Growth	0.039***	0.179***	0.115***	0.048***	0.069***	0.008	0.003	0.293***	0.018	0.034***	0.004	-0.0142***	1.000		
ROA	-0.068***	0.571***	0.257***	0.047***	0.078***	0.187***	0.014	0.165***	-0.253***	0.108***	0.174***	-0.071***	0.287***	1.000	
PPE	-0.231***	-0.070***	-0.031***	-0.003	0.014	0.186***	-0.046***	-0.136***	0.083***	0.089***	0.172***	0.103***	-0.120***	-0.029***	1.000
VIF	1.11	-	1.22	1.24	1.21	1.77	1.03	1.35	1.42	2.03	1.47	1.32	1.22	1.36	1.12
Mean vif	1.36														

p-values in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5
Regression results.

Variables	Model (1) Baseline	Model (2) DW × TE	Model (3) DW × KV	Model (4) DW × Attention
	SG	SG	SG	SG
primary variables				
DW	−7.361** (3.599)	−7.638** (3.625)	−7.696** (3.631)	−7.450** (3.551)
TE		0.024 (0.023)		
DW*TE		−35.604** (17.787)		
KV			0.010** (0.004)	
DW*KV			−15.790** (7.742)	
Attention				0.003*** (0.001)
DW*Attention				−5.049*** (1.437)
Control Variables				
Growth	−0.018*** (0.003)	−0.018*** (0.002)	−0.018*** (0.003)	−0.018*** (0.003)
ROA	0.506*** (0.035)	0.505*** (0.035)	0.504*** (0.034)	0.505*** (0.034)
PPE	−0.035*** (0.011)	−0.035*** (0.011)	−0.035*** (0.011)	−0.035*** (0.011)
Inst	0.018 (0.014)	0.018 (0.014)	0.017 (0.014)	0.019 (0.014)
Indp_z	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
PB_z	0.010*** (0.001)	0.010*** (0.001)	0.010*** (0.001)	0.010*** (0.001)
Lev	0.006 (0.012)	0.006 (0.012)	0.006 (0.012)	0.006 (0.012)
Size	−0.005*** (0.002)	−0.005*** (0.002)	−0.005*** (0.002)	−0.005*** (0.002)
Age_z	0.017*** (0.002)	0.017*** (0.002)	0.017*** (0.002)	0.017*** (0.002)
Top1	−0.067*** (0.011)	−0.065*** (0.011)	−0.066*** (0.011)	−0.065*** (0.011)
Constant	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)
FE Year	Yes	Yes	Yes	Yes
Observations	7564	7564	7564	7564
R-squared	0.67	0.67	0.67	0.67
F	57.73	64.17	48.94	51.86

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

pattern aligns with its nature as a long-term sustainability indicator. Such indicators typically exhibit limited short-term volatility. To assess its economic relevance, we translate the estimated coefficient into economically meaningful magnitudes. Utilizing the interquartile range of digital washing in the regression sample ($DW(P25) = 0.000026$; $DW(P75) = 0.000231$), moving from the 25th to the 75th percentile of DW implies a decrease in SG of 0.001512. Relative to the sample dispersion of SG ($SD = 0.061601$), this corresponds to -0.0245 standard deviations. By employing a broader percentile contrast ($P10-P90$), the implied change in SG is -0.004225 . This is equivalent to -0.0686 standard deviations. These magnitudes indicate that the negative impact of digital washing on sustainable growth is not only statistically significant but also economically meaningful.

4.3.2. Empirical analysis results

The interaction coefficients β_3 for $DW \times TE$, $DW \times KV$, and $DW \times Attention$ are all negative and significant. This indicates that higher levels of the moderators strengthen the negative marginal effect of digital washing on corporate sustainable growth. This is consistent with the monotonic decline of simple slopes from -1σ to $+1\sigma$ in Fig. 1. It is also reflected by the steepest fitted line at $+1\sigma$ in Fig. 2.

As shown in Table 5 Model (2) $DW \times TE$, the regression coefficient of

the interaction term $DW \times TE$ is significantly negative. The result is significant at the 5% level ($\beta = -35.604$, $p < 0.05$). This indicates that the higher the technical efficiency, the stronger the inhibiting effect of digital washing on the sustainable growth of enterprises. The statistical results are robust. From a theoretical standpoint, enterprises with high technical efficiency possess greater substantive digitalization and innovation potential. Technological innovation is supposed to promote corporate sustainable development (Omonijo and Zhang, 2025). However, utilizing high technical efficiency for “digital washing” may lead to wasted potential and resource misallocation. Therefore, in technologically advanced firms, digital washing is more easily identified as false advertising. This significantly weakens sustainable growth.

As shown in Table 5 Model (3) $DW \times KV$, the regression coefficient of the interaction term $DW \times KV$ is significantly negative, with results being significant at the 5% significance level ($\beta = -15.790$, $p < 0.05$). This indicates that higher quality of information disclosure leads to a more pronounced negative impact of digital washing on corporate sustainable growth, and the statistical results are robust. This indicates that higher-quality information disclosure leads to a more pronounced negative impact of digital washing on corporate sustainable growth. The statistical results are robust. This finding is consistent with the theory of information asymmetry. He and Ismail (2024) find that comprehensive ESG information disclosure significantly reduces information asymmetry between enterprises and investors. When the disclosure level is high, it becomes difficult for enterprises to obscure digital washing behavior through ambiguous publicity, making it easier for investors and regulators to detect discrepancies between corporate actions and stated commitments, exerting stronger pressure and imposing more severe penalties on enterprises. Thus, digital washing is less effective in a high-disclosure environment, further exacerbating its inhibitory effect on the corporate sustainable growth.

As shown in Table 5 Model (4) $DW \times Attention$, the regression coefficient of the interaction term $DW \times Attention$ is significantly negative, with results being significant at the 1% significance level ($\beta = -5.046$, $p < 0.01$). This indicates that as investor attention increases, the suppressive effect of digital washing on the sustainable growth of enterprises becomes stronger. The statistical results are robust. Zhang et al. (2022) find that investor attention can significantly enhance the ESG performance of listed companies. High attention implies that market participants exercise more stringent oversight of corporate behavior. This makes it easier for digital washing to be identified and elicits negative reactions from investors. With enhanced investor attention, the market's expectations of companies' commitment to digitalization and corporate sustainable growth also increase. False digital claims are more likely to be scrutinized and challenged. This exerts a more significant negative impact on sustainable growth.

5. Robustness check

We using peer digital washing from the same industry-year but different provinces as an instrument. However, the first-stage results indicate that peer DW has limited explanatory power for firm-level digital washing. This suggests that DW is largely driven by firm-specific strategic behavior rather than industry-level diffusion. Therefore, we do not rely on the IV estimates as the primary identification strategy. Accordingly, we adopt a more conservative interpretation of our findings. We focus on the consistent negative association between digital washing and corporate sustainable growth documented across fixed-effects regressions, alternative DW constructions, PSM analyses, and dynamic specifications.

5.1. Endogeneity diagnostics and IV-2SLS estimation

To alleviate potential endogeneity issues, this study employs the two-stage least squares (2SLS) method for robustness checks. Following existing literature (Angrist and Krueger, 2001; Wooldridge, 2002), we

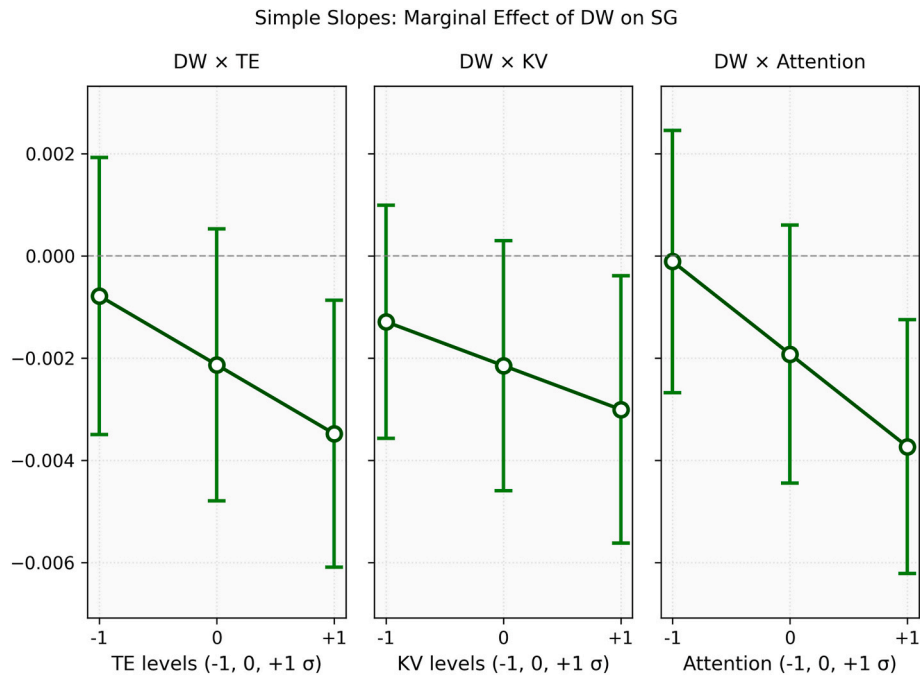


Fig. 1. Simple slopes of DW on SG at low (-1σ), mean (0), and high ($+1\sigma$) levels of the moderator. Panel A: TE; Panel B: KV; Panel C: Attention. Dots denote marginal effects with 95% CIs. Model specifications follow Table 5.

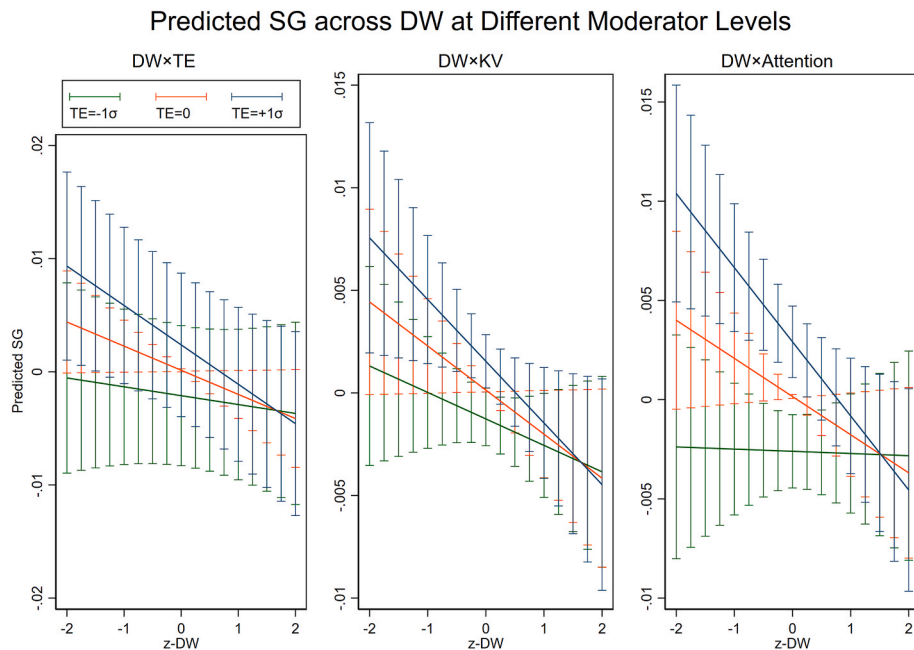


Fig. 2. Predicted sustainable growth (SG) at different levels of digital washing (DW). The figure plots predicted values of SG across low (-1σ), mean (0), and high ($+1\sigma$) levels of DW, based on the estimated coefficients. Shaded areas represent 95% confidence intervals. Model specifications follow Table 5.

construct an interaction term between the company's price-to-earnings ratio (PE) and the industry average investor attention as the instrumental variable. The specific selection criteria are as follows: (1) Firms with a high price-to-earnings ratio and strong industry-level investor attention. These firms typically face greater scrutiny from capital markets. This scrutiny makes their digital narratives and disclosure practices more susceptible to external examination, which significantly impacts digital washing (Stock and Yogo, 2002). Therefore, this interaction term has a strong correlation with digital washing and satisfies the relevance condition for instrumental variables. (2) Industry-level investor

attention and valuation expectations do not directly determine corporate sustainable growth performance. Their impact is more reflected in promoting digital information disclosure and symbolic responses, rather than directly affecting the fundamental operations of the firm (Liu and Zhao, 2025; Suchman, 1995a,b). Thus, this instrumental variable influences corporate sustainable growth performance solely through its effect on digital washing, satisfying the exclusion restriction.

As shown in Table 6 (1), the results of the first stage indicate that the instrumental variable significantly explains DW. The Kleibergen–Paap test ($LM = 16.657$, $p < 0.001$) rejects the null hypothesis of under-

Table 6

Robustness check results.

	(1)	(2)	(3)	(4)
	2SLS regression		SPM	Dynamic Effect
	First Stage	Second Stage	SG	Lagged SG
DW		−44.389** (20.007)		−9.677* (4.870)
Instrumental Variable	−0.600*** (0.009)			
DW_SPM			−7.361** (3.599)	
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Obs.	8237	8237	7564	4679
Adj.R2	/	0.394	0.574	0.546
Kleibergen–Paap rk LM statistic rk LM statistic	16.657 [p = 0.000]			
Kleibergen–Paap rk Wald F	45.137			
χ^2 (1)	4.763 [p = 0.0291]			

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include firm and year fixed effects, with control variables as listed in the table.

identification; the Kleibergen–Paap rk Wald weak instrument test yields a result of $F = 45.137$, which is well above the critical value of 16.38 under Stock–Yogo's 10% maximal bias, alleviating concerns regarding weak instruments. The endogeneity test ($\chi^2 = 4.763$, $p = 0.0291$) significantly rejects the null hypothesis of “DW exogeneity,” indicating that the use of IV–2SLS is necessary. The results of the second stage show that the coefficient for DW is -44.389 ($p < 0.05$), suggesting that the test results are consistent with the baseline regression direction, and that digital washing significantly suppresses the outcomes of corporate sustainable growth (see Table 7).

5.2. Propensity score matching (PSM)

To further validate the robustness of the empirical research conclusions and effectively mitigate endogeneity issues and potential sample

selection bias, this paper employs the 0.05 caliper propensity score matching (PSM) method for robustness testing. First, the sample is divided into two groups based on the median of DW. Enterprises above the median are defined as the treatment group ($DW_SPM = 1$), while those below the median are defined as the control group ($DW_SPM = 0$). Subsequently, a Logit model is used to estimate the propensity score for enterprises entering the high digital washing group. The nearest neighbor 1:1 matching method is then applied, ultimately resulting in 7564 valid matched samples.

To assess the covariate balance after propensity score matching (PSM), this study employs the Standardized Mean Difference (SMD) along with several accompanying diagnostic indicators. The SMD is a key metric for evaluating whether the distribution of covariates between the treatment and control groups approaches balance. It effectively mitigates the influence of scale and is widely used for PSM balance checks. The results indicate that the standardized mean deviation of the samples before matching is 22.9%. This value significantly decreases to 2.8% after matching, falling below the commonly used threshold of 10%. This suggests a significant convergence of systematic differences among covariates. Furthermore, the Pseudo R^2 declines from 0.080 to 0.002. The p-value of the joint significance test becomes non-significant post-matching. This indicates that the covariates no longer jointly explain group membership in the treatment group, thus reflecting good matching quality. Additionally, the variance ratios of all covariates fall within the acceptable range of [0.50, 2]. It is evident that PSM significantly improves the covariate balance between groups. This effectively reduces selection bias and thereby enhances the credibility of causal inferences.

Based on this, we further conducted an OLS regression on the matched samples to examine the causal relationship between digital washing and corporate sustainable growth. As shown in Table 6(3), the regression results for DW_SPM ($\beta = -7.361$, -0.006 , $p < 0.05$) are consistent with the baseline regression. This indicates that even after effectively mitigating issues of endogeneity and sample selection bias, the digital washing behavior of enterprises still has a significant negative impact on corporate sustainable growth. This further enhancing the robustness and credibility of the research conclusions.

5.3. Dynamic inspection

To further validate the reliability of the conclusions, this paper introduces the lagged value of the dependent variable (Lagged SG) as an alternative indicator for dynamic regression testing. The underlying

Table 7

Balance test results after propensity score matching.

Variable	Sample	Treated Mean	Control Mean	%bias	%Bias Reduction	t-stat	p-value	V(T)/V(C)
Inst	U	−3.071	6.721	−39.1		−17.730	0.000	1.000
	M	−3.082	−3.337	1.0	97.40%	0.460	0.646	0.980
Indp	U	0.282	−0.202	9.0		4.100	0.000	1.020
	M	0.280	0.619	−6.3	29.90%	−2.760	0.006	0.88*
PB	U	0.412	−0.115	17.5		7.920	0.000	1.060
	M	0.413	0.555	−4.7	72.90%	−1.950	0.051	0.74*
Lev	U	−0.021	0.010	−15.9		−7.210	0.000	0.88*
	M	−0.020	−0.020	−0.2	98.60%	−0.100	0.922	0.85*
Size	U	−0.137	0.041	−15.5		−7.050	0.000	0.82*
	M	−0.136	−0.120	−1.4	90.80%	−0.680	0.499	0.950
Top1	U	−1.675	4.467	−40.1		−18.220	0.000	0.82*
	M	−1.688	−1.625	−0.4	99.00%	−0.200	0.844	0.950
Age	U	−0.786	0.786	−20.9		−9.480	0.000	0.84*
	M	−0.788	−1.012	3.0	85.80%	1.410	0.159	1.000
Growth	U	0.011	−0.011	9.6		4.340	0.000	1.45*
	M	0.011	0.018	−3.5	63.70%	−1.420	0.155	0.93*
ROA	U	−0.003	0.003	−10.6		−4.830	0.000	1.24*
	M	−0.003	0.000	−5.0	44.97%	−2.300	0.022	0.970
PPE	U	−0.033	0.033	−51.2		−23.240	0.000	0.58*
	M	−0.033	−0.031	−2.1	96.00%	−1.120	0.263	1.07*

The asterisk in V(T)/V(C) indicates variance ratios outside the recommended range [0.50, 2].

logic is based on two considerations. First, from a theoretical perspective, digital narratives do not immediately and fully reflect in corporate sustainable growth. Instead, they require processes such as resource reallocation, green R&D cycles, and updates in external market perceptions (Lin et al., 2025; Quttainah and Ayadi, 2024; Song et al., 2023; Yoshikuni et al., 2024). Second, from a measurement perspective, employing a lagged dependent variable is a common distributed lag setup. This approach effectively identifies intertemporal effects, mitigates biases caused by synchrony, and aligns with the treatment methods of classical distributed lag models (Almon, 1965; Klein, 1955). The results presented in Table 6 (4) indicate that the regression coefficient of digital washing remains significantly negative in the lagged period ($\beta = -9.677$, $p < 0.1$). This is consistent with the baseline regression results. This finding suggests that even when accounting for the dynamic effects of corporate sustainable growth, the inhibitory effect of digital washing on corporate sustainable growth persists and does not diminish over time. Consequently, these results help to mitigate the influence of short-term shocks or endogeneity issues. This further validates the robustness of the conclusions drawn in this paper.

5.4. Placebo test

To further assess the causal direction of the relationship between digital washing and corporate sustainable growth, we conduct a series of placebo tests by replacing the contemporaneous digital washing measure with its future values. Specifically, we sequentially include one-period, two-period, and three-period leads of digital washing (DW_{t+1} , DW_{t+2} , and DW_{t+3}) as explanatory variables. We maintain the same set of control variables and fixed effects.

As reported in Table 8, the coefficient remains statistically significant. This potentially reflects the persistence of digital narrative strategies and the gradual nature of sustainable growth outcomes. Notably, when extending the placebo window to more distant future values, the estimated coefficients become statistically insignificant. This pattern suggests that future digital washing does not systematically predict current sustainable growth. Collectively, these placebo results imply that the baseline findings are unlikely to be influenced by reverse causality or anticipation effects. The negative association between digital washing and sustainable growth primarily operates contemporaneously. This supports the causal interpretation of the main regression results (see Table 9).

5.5. Alternative measures of digital washing

To examine the robustness of our findings regarding alternative operationalizations of digital washing, we construct relative measures by re-scaling the baseline digital washing indicator within industry-year cells. Specifically, we compute (i) an industry-year standardized score (DW_z) and (ii) an industry-year percentile rank (DW_prank). These measures capture firms' relative positions in the distribution of digital washing among peer firms operating in similar technological and institutional environments.

The results are reported in Table 8. In contrast to the baseline specification, the coefficients for both DW_z and DW_prank are statistically insignificant. This finding suggests that the adverse effect on sustainable growth is not driven by firms' relative rankings in digital washing within a given industry-year. Instead, it is driven by the absolute extent of excessive digital disclosure relative to firm fundamentals. In other words, digital washing appears to have significant consequences when firms substantially deviate from their own underlying digital capabilities. This is different from merely differing from peers in relative terms. Taken together, these results further clarify the construct boundary of digital washing. They also reinforce the interpretation that the baseline measure captures economically meaningful excess disclosure rather than industry-level or relative disclosure patterns.

6. Heterogeneity analysis

Differences in firm size and management team size can be understood through the lens of organizational complexity. Larger firms and management teams encounter heightened coordination costs and information frictions. These factors may increase the likelihood that digital narratives serve as substitutes for substantive strategic adjustments. Consequently, excessive digital disclosure is more likely to undermine effective resource allocation and organizational learning. This, in turn, amplifies its adverse effects on sustainable growth.

As shown in Table 8, models (1) and (2) classify the samples into service-oriented enterprises and non-service-oriented enterprises. The results indicate that the regression coefficient for digital washing in service-oriented enterprises is significantly negative. The result is significant at the 5% significance level ($\beta = -8.376$, $p < 0.05$). This suggests that digital washing has a significant inhibiting effect on the corporate sustainable growth of service-oriented enterprises. However, in non-service-oriented enterprises, the coefficient is not significant. This indicates that the negative impact is not pronounced in non-service-

Table 8
Placebo tests and Alternative DW measures.

	Main effects	Placebo 1	Placebo 2	Placebo 3	industry-year	re-scaling
	SG	SG	SG	SG	SG	SG
DW	-7.361** (3.599)					
DW_lead1		-9.579** (3.671)				
DW_lead2			-8.465 (5.558)			
DW_lead3				-0.449 (4.783)		
DW_z					-0.001 (0.001)	
DW_prank						-0.001 (0.002)
Control	YES	YES	YES	YES	YES	YES
FE Year	YES	YES	YES	YES	YES	YES
R-squared	0.6712	0.7069	0.7373	0.7551	0.6722	0.6721
F	57.73	52.21	22.12	21.69	54.50	54.90
Constant	0.001* (0.001)	0.005*** (0.001)	0.007*** (0.002)	0.005*** (0.001)	-0.000 (0.000)	0.000 (0.001)
Observations	7564	5403	3808	2670	7432	7432

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9
Heterogeneity analysis regression results.

	(1)	(2)	(3)	(4)	(5)	(6)
	Service Enterprises	Non-Service Enterprises	Large Enterprises	Small Enterprises	Large management teams	Small management teams
DW	-8.376*** (-4.619)	-7.310 (-1.633)	-12.008*** (-2.669)	3.399 (0.728)	-8.579* (-1.776)	-6.166 (-1.347)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.511	0.589	0.716	0.400	0.623	0.572
Observations	1000	6554	3317	3363	3000	4030

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

oriented enterprises. This finding indicates that service enterprises are more vulnerable to the adverse effects of digital washing during their digital transformation. On one hand, service enterprises heavily rely on external reputation and market trust. If the practice of excessively exaggerating digital achievements is revealed, it can easily undermine the trust of investors and customers. This directly diminishes the market competitiveness and long-term value of the enterprise. On the other hand, the service industry inherently lacks substantial asset backing. This makes it more susceptible to reputational risks stemming from misleading narratives. These risks can be rapidly amplified, consequently exacerbating the inhibitory effects on corporate sustainable growth.

As illustrated in Table 8 models (3) and (4), the regression analysis grouped by enterprise size shows that the inhibitory effect of digital washing on large enterprises is significantly negative. The result is significant at the 1% level ($\beta = -12.008$, $p < 0.01$). This indicates that the negative impact of digital washing on the corporate sustainable growth of large enterprises is more pronounced. Conversely, in small enterprises, the coefficient is not significant. This suggests that the inhibitory effect is not significant in small enterprises. This finding suggests that large enterprises, due to increased scrutiny from the capital market, investors, and the media, are more likely to have misleading elements in their digital narratives exposed. This leads to negative market reactions.

Furthermore, large enterprises typically face greater social responsibilities and compliance pressures. These factors make their digital washing practices more susceptible to scrutiny by regulatory bodies or the public. This, in turn, exacerbates the inhibitory effects on corporate sustainable growth.

As shown in Table 8, Models (5) and (6) classify firms into groups based on the size of their management teams: large and small. The results indicate a significantly negative coefficient for digital washing in firms with larger management teams ($\beta = -8.579$, $p < 0.10$). The coefficient for firms with smaller management teams is statistically insignificant. This pattern suggests that the adverse effect of digital washing on corporate sustainable growth is more pronounced in organizations with larger management teams. This heterogeneity can be understood from the perspective of internal governance and coordination. Larger management teams are often associated with fragmented decision authority, diluted accountability, and increased internal coordination costs. Under these conditions, digital narratives may serve as a low-cost symbolic alignment mechanism among executives. This makes substantive strategic adjustments and operational follow-through more challenging. Consequently, digital washing is more likely to replace genuine digital capability building and sustainability-oriented actions in firms with larger management teams.

In contrast, smaller management teams typically exhibit clearer lines

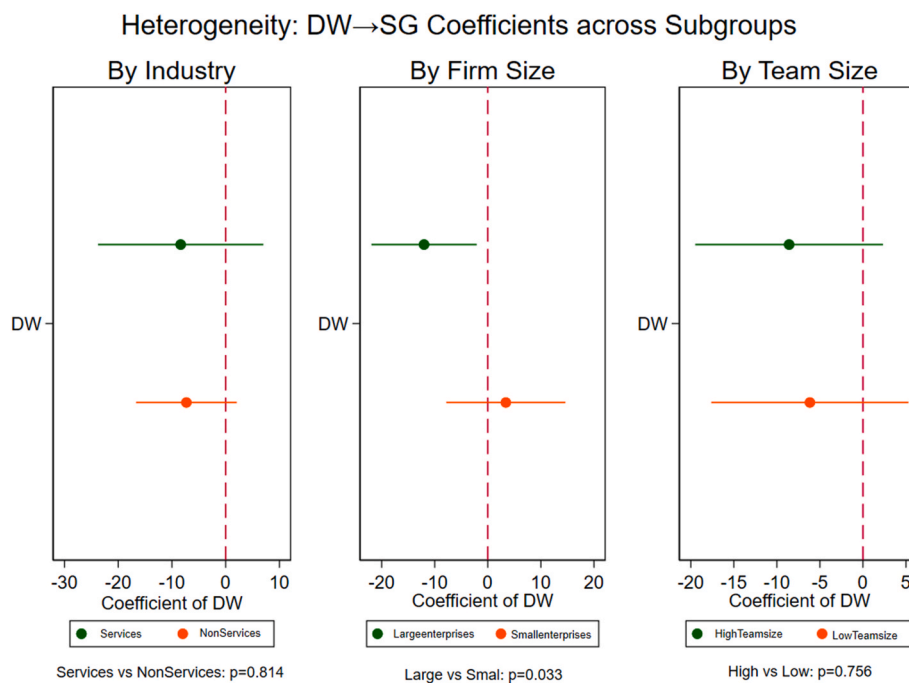


Fig. 3. Heterogeneity: DW→SG coefficients across subgroups. Panel A: Industry (Services vs Non Services); Panel B: Firm size (Large enterprises vs Small enterprises); Panel C: Team size (High Team size vs Low Team size). Dots denote the DW coefficients from subgroup regressions; horizontal bars indicate 95% confidence intervals; the vertical dashed line marks zero.

of responsibility, more efficient internal communication, and stronger alignment between strategic intent and execution. These characteristics facilitate closer integration between digital initiatives and substantive innovation or sustainability practices. This helps mitigate the adverse effects of digital washing on sustainable growth.

As shown in Fig. 3, the heterogeneity of the digital washing effect is visually reflected across different subgroups. Specifically, Panel A compares service and non-service industries. Panel B distinguishes between large and small enterprises. Panel C contrasts firms with high versus low team sizes. The dots represent the estimated coefficients, while the horizontal bars indicate 95% confidence intervals. These visual results are consistent with the subsequent regression tables. Evidently, the inhibitory effect of digital washing on corporate sustainable growth is more pronounced in the service industry, among large enterprises, and within organizations characterized by high team sizes. This finding further underscores the variability and context-dependency of the impact of digital washing. It suggests that regulatory bodies and enterprise managers should consider key factors such as industry, scale, and governance structure when formulating digital strategies and governance measures. This approach aims to enhance the authenticity and effectiveness of digital strategies and avoid the development dilemma of prioritizing form over substance.

It is important to note that differences in statistical significance across subsamples may partially reflect variations in sample size rather than fundamental differences in the underlying economic effects. Specifically, some subsamples are characterized by smaller numbers of observations, which can reduce statistical power and attenuate significance levels. To address this concern, we complement significance-based inference with effect size measures. This is done by translating coefficient estimates into economically meaningful magnitudes, such as interquartile and percentile-based changes. The results indicate that even when statistical significance varies across subsamples, the implied economic effects remain consistent in direction and magnitude. This suggests that the observed heterogeneity is not solely driven by sample imbalance.

7. Discussion and conclusion

7.1. Discussion

Digital washing undermines firms' corporate sustainable growth ($\beta = -7.361$, $p < 0.05$). This suggests a long-run governance cost of symbolic digital narratives. Prior studies often stress the benefits of substantive digital transformation (Orlikowski and Barley, 2001; Vial, 2021). Our findings qualify this view. Digital rhetoric without capability building can weaken sustainable growth. Whereas prior studies generally report that substantive digital transformation improves firms' environmental performance, innovation, and long-term value creation (Orlikowski and Barley, 2001; Vial, 2021; Luo and Liu, 2024), our evidence highlights an important boundary condition: digital narratives can be strategically decoupled from operational change. By isolating "digital washing"—excessive digital rhetoric relative to a reasonable benchmark—we show that the sustainability dividend of digitalization is not automatic. This distinction helps reconcile mixed findings in the digitalization-sustainability literature, suggesting that positive effects are more likely when digital disclosures are aligned with genuine capability building.

The negative effect is stronger when firms have higher technical efficiency, better disclosure quality, or greater investor attention. This differs from the greenwashing literature. There, transparency and external oversight often mitigate negative outcomes (Cao et al., 2022; Zhang et al., 2023). Legitimacy theory also predicts that scrutiny should push firms toward substantive change (Suchman, 1995a,b). We observe the opposite in digital washing. Pressure can incentivize more strategic digital storytelling. This mechanism distinguishes digital washing from traditional greenwashing. Moreover, the present study extends the

emerging digital washing literature beyond short-term outcomes. Prior work has mainly examined market reactions or accounting attributes (Wei et al., 2025). Our results indicate that the consequences of symbolic digital narratives are not transient; they accumulate and manifest in firms' sustainable growth capacity, implying that digital washing is a strategic governance distortion that crowds out substantive capability accumulation and long-term value creation.

This does not imply that disclosure quality or external governance is ineffective. The effect is conditional. When disclosure is decoupled from capability building, transparency increases verifiability. It raises stakeholder expectations. The gap between narrative and performance becomes easier to detect. This accelerates reputational loss and governance backlash. Therefore, high-quality disclosure can amplify the costs of digital washing. These moderators are relatively stable. They condition the impact of digital washing.

Importantly, the amplifying roles of technical efficiency, disclosure quality, and investor attention reveal a "credibility trap" in digital settings. In high-capability or high-transparency firms, stakeholders' prior beliefs about digital competence are stronger, and digital narratives diffuse more widely. When a firm engages in digital washing, this creates a larger expectation gap between narratives and observable outcomes. Once inconsistencies are detected—often with a lag due to the opacity and long implementation cycles of digital initiatives—the reputational and governance backlash can be sharper. This mechanism differs from traditional greenwashing, where many environmental outcomes are more directly observable and monitoring more often constrains symbolic behavior (Cao et al., 2022; Zhang et al., 2023).

Overall, our empirical evidence yields three unique insights. First, we provide large-sample evidence that digital washing, distinct from substantive digital transformation, undermines long-term corporate sustainable growth. Second, we identify counterintuitive boundary conditions: operational capability (technical efficiency), transparency (disclosure quality), and external scrutiny (investor attention) can amplify—not buffer—the adverse effects of symbolic digital narratives. Third, the heterogeneity results show that these effects are most pronounced in service industries, large firms, and organizations with larger management teams, offering actionable screening cues for regulators, investors, and boards.

To reinforce the reliability of our findings, we conducted several robustness assessments. These included two-way fixed-effects estimation, IV-2SLS identification, propensity score matching, dynamic lagged panel regressions, and a variety of heterogeneity tests. All results converge in the same direction, reaffirming the long-term suppressive impact of digital washing on corporate sustainable growth.

Heterogeneity analyses further show that the effect varies across contexts. It is stronger in the service sector, where outputs are more intangible and harder to verify, increasing reliance on narratives and information asymmetry. The negative effect is also more pronounced in large firms and in firms with larger management teams. These settings typically face higher public and market scrutiny and greater internal coordination costs, which can increase the incentives and opportunities for symbolic digital narratives and resource misallocation.

7.2. Conclusion

This study investigates the governance consequences of digital washing by examining its association with corporate sustainable growth. The analysis utilizes a panel of Chinese A-share listed firms from 2014 to 2023. In this study, SG is conceptualized as a firm's long-term capacity to maintain stable economic growth while fostering environmental responsibility and creating social value within the confines of governance and strategic constraints. Consistent with established sustainability frameworks, SG encompasses long-term developmental outcomes rather than short-term performance fluctuations. Using a residual-based textual measure and a two-way fixed-effects model, we find that digital washing is negatively associated with corporate sustainable growth. The

result is robust across alternative specifications and identification tests. The negative link is stronger when firms have higher technical efficiency, higher disclosure quality, and higher investor attention. These moderators highlight the role of capability and information environments.

By distinguishing symbolic digital narratives from substantive digital transformation, this study contributes to the digitalization–sustainability literature in three ways: (1) it develops a reproducible, residual-based text metric of digital washing; (2) it provides large-sample evidence of the long-term governance cost of digital washing for corporate sustainable growth; and (3) it uncovers boundary conditions showing that capability, transparency, and attention can intensify the penalties of symbolic digitalization.

7.3. Revelation

7.3.1. Theoretical contributions

Compared with prior studies on substantive digital transformation (Orlikowski and Barley, 2001; Vial, 2021) and the emerging digital washing evidence focused on short-horizon accounting outcomes (Wei et al., 2025), we link digital washing to long-term sustainable growth and clarify the conditions under which its governance costs are most severe.

This study contributes to research on digital transformation and corporate sustainable growth. First, we articulate a clear definition of digital washing and differentiate it from substantive digital transformation. By embedding this concept within an institutional pressure–impression management framework, we explain why firms may exhibit worsening corporate sustainable growth outcomes despite adopting similar digital narratives. Second, we introduce and validate an innovative three-dimensional moderation model. The results indicate that factors typically viewed as performance-enhancing—such as strong technological efficiency, robust disclosure practices, and heightened external scrutiny—can significantly amplify the negative consequences of digital washing. This shifts the scholarly conversation from assessing whether digital washing has effects to understanding when and under what conditions these effects intensify. It thereby extends legitimacy and decoupling theories by identifying boundary conditions for symbolic behavior. Third, by incorporating resource misallocation into the analytical framework, we connect digital washing to long-term corporate performance. Rather than focusing solely on short-term market reactions, we demonstrate how symbolic digital actions displace substantive investment, weaken future value creation, and influence trajectories of capability development. Finally, using evidence from China's institutional context, this study addresses a notable gap in research on emerging markets. It reveals how digital washing unfolds under China's distinct regulatory and market configurations. It also contributes a comparative lens for global discussions on corporate legitimacy and digital strategy.

7.3.2. Methodological contributions

This study advances the methodology for examining digital washing in several significant ways. First, we develop a replicable text-analytic metric of digital washing based on firms' annual report disclosures. We also incorporate critical moderators—including technological efficiency, disclosure robustness, and investor attention—into the empirical framework. This combined approach yields a systematic identification process. It can detect embellished digital narratives and assess their organizational consequences. Second, in contrast to earlier work that largely relies on isolated event-based analyses, our design represents a shift towards a more comprehensive, data-driven assessment of digital strategy authenticity. This methodological shift provides a transferable template. Future studies can adapt and refine this template across various institutional and industrial contexts.

7.3.3. Practical implications

For managers, our findings underscore the risks associated with pursuing digital washing as a short-term tactic. Executives should resist the temptation to rely on catchy yet hollow digital narratives and instead refocus on the fundamental objective of digital transformation—developing genuine technological capabilities. This requires sustained investment in infrastructure, workforce skills, and data-driven decision-making. These efforts ensure that digital initiatives lead to measurable performance improvements. While polished digital narratives may temporarily generate market enthusiasm, they ultimately undermine long-term credibility and value. To mitigate this risk, firms should strengthen internal review processes, ensure the accuracy and transparency of digital disclosures, and avoid exaggerated claims. These measures will help ensure that digital strategies contribute to authentic and sustainable growth while protecting the firm's reputation.

For regulators, the findings offer clearer guidance on how to identify and deter digital washing in supervisory practice. Our evidence indicates that digital washing is typically manifested in a pronounced divergence between firms' intensive digital-related narratives and weak or absent improvements in efficiency and long-term growth outcomes. Such a pattern provides a useful risk signal for regulatory scrutiny, particularly when digital disclosures expand rapidly while outcome-oriented indicators remain stagnant or deteriorate. From a deterrence perspective, the results suggest that regulatory approaches emphasizing the consistency between digital claims and verifiable performance outcomes can help curb digital washing. By placing greater weight on whether digital narratives are supported by observable operational and growth-related evidence, regulators can reduce the benefits of symbolic digital communication and increase the costs of exaggerated digital rhetoric. In this process, information generated by auditors' assurance activities or ESG-related assessments may serve as complementary reference inputs. However, the core regulatory focus should remain on aligning digital disclosures with substantive economic outcomes.

For investors and analysts, this study underscores the need for more rigorous due diligence to uncover inflated digital claims. Instead of accepting firms' self-reported digital successes at face value, investors should scrutinize substantive indicators—such as R&D intensity, patent output, and the depth of information systems implementation—to assess whether genuine digital transformation is occurring. External sources, including independent industry reports, third-party ratings, and AI-driven text analytics, can help validate disclosures and identify inconsistencies or exaggerations. Integrating digital governance into ESG assessment frameworks and prioritizing transparent, verifiable digital outcomes can mitigate the risk of being misled by superficial digital behaviors. This vigilance not only protects investor interests but also motivates firms to pursue authentic, rather than performative, digital innovations.

7.4. Limitations and future research directions

While this study provides compelling evidence on how digital washing undermines corporate sustainable growth, several limitations warrant acknowledgment. First, this study draws exclusively on Chinese A-share listed firms. Our evidence primarily reflects how digital washing operates within China's particular institutional arrangements and cultural norms. In advanced economies, investor protection tends to be stronger, disclosure rules are more rigorously enforced, and capital markets function with higher transparency. Under such conditions, digital-related disclosures are typically examined through more stringent auditing and regulatory procedures. These mechanisms may narrow the room for digital washing or push it into more subtle forms. In contrast, emerging markets often face weaker institutional safeguards and uneven development in digital infrastructure and supervisory capacity. Firms in these settings may rely more heavily on symbolic digital narratives to offset institutional voids. This reliance can increase the likelihood of digital washing and intensify its impact on resource-

allocation distortions and reputational risk. Cultural differences across countries also play an important role. Variations in power distance, collectivism, and uncertainty avoidance shape how external stakeholders interpret corporate digital narratives and how tolerant they are toward symbolic forms of compliance. These cultural factors can meaningfully alter the observable impact of digital washing on corporate sustainable growth outcomes. Taken together, the institutional and cultural heterogeneity across national contexts suggests that the external validity of our findings should be considered cautiously when applied beyond China. Furthermore, our measurement of digital washing is based on keyword-driven text mining. Although this represents a methodological advancement, it may not fully capture contextual nuances such as negation, irony, or sector-specific terminology. Additionally, firms may articulate similar behaviors utilizing varied expressions, which could introduce identification bias. Proxy variables, such as search trend indices employed to gauge investor attention, may also be susceptible to noise or distortions at the platform level, thereby affecting robustness. A further limitation pertains to the ethical implications of data-driven research. Approaches that utilize large-scale data and AI systems must prioritize privacy safeguards and algorithmic fairness. Ensuring transparency and responsible data usage should be a fundamental focus in future studies.

Looking ahead, several promising research directions emerge. First, expanding the dataset to include firms from multiple countries would facilitate cross-national comparisons. This approach would help assess whether digital washing measures remain stable across different contexts and reveal how institutional and cultural conditions shape their effects. It would thereby enhance the global relevance of the findings. Second, future research could explore more sophisticated analytical tools. Transformer-based NLP models or machine learning techniques may more effectively identify semantic nuances in corporate

communications. Combining textual measures with observable operational indicators—such as R&D spending, patenting, or technology adoption—can improve accuracy. Methods like anomaly detection or causal inference may further strengthen efforts to pinpoint digital washing and its consequences. Third, a greater micro-level investigation is necessary. Case studies, executive interviews, experimental designs, or longitudinal tracking can illuminate how digital narratives translate into internal resource allocation and long-term performance. Such approaches can also reveal how governance structures and managerial practices either exacerbate or mitigate symbolic digital behavior. They can clarify how firms balance external digital storytelling with internal technological capability building. By pursuing these directions, future research can deepen our understanding of digital washing and inform more effective governance mechanisms. This will support digital transformation that delivers genuine, rather than performative, impact.

CRediT authorship contribution statement

Lijun Wang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Shanyue Jin:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

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Appendix

Appendix A. The distributional properties of the digital washing (DW) measure

Figure A1a presents the distribution of the raw residuals ($\hat{e}$) obtained from the digital disclosure model before truncation. The residuals are centered around zero and display a symmetric shape with thin tails, indicating that the first-stage model is well specified and does not exhibit systematic bias.

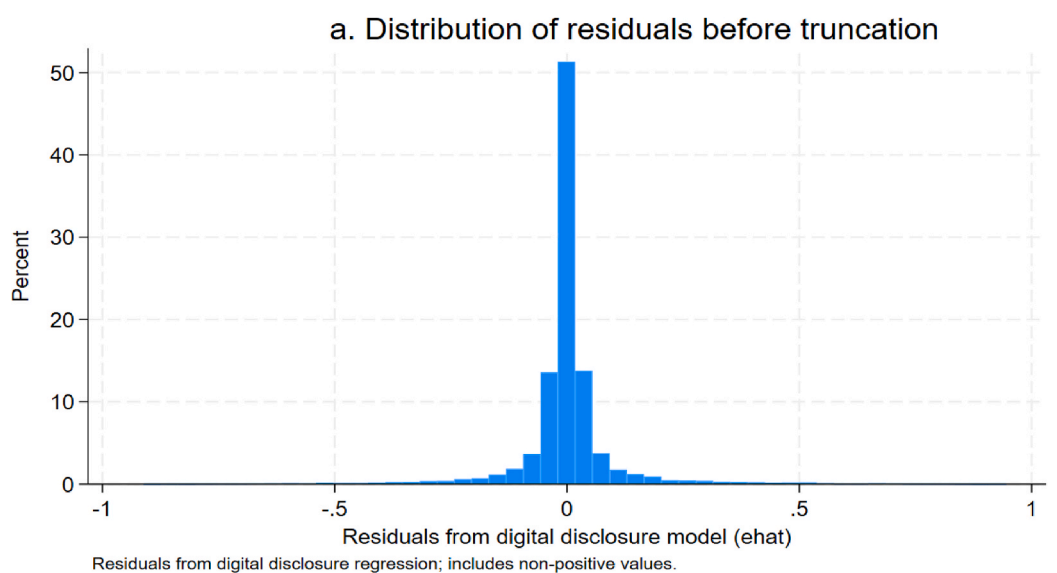


Fig. A1a. Distribution of residuals before truncation

Figure A1b illustrates the distribution of digital washing (DW), defined as the positive residuals only ($DW > 0$), which constitute the regression

sample. As anticipated due to its residual-based and truncated construction, DW displays a highly right-skewed distribution, with the majority of observations concentrated at relatively low levels and a long right tail representing firms with significant symbolic overstatement of digitalization. This pattern reflects the conceptual design of DW rather than a loss of measurement sensitivity.

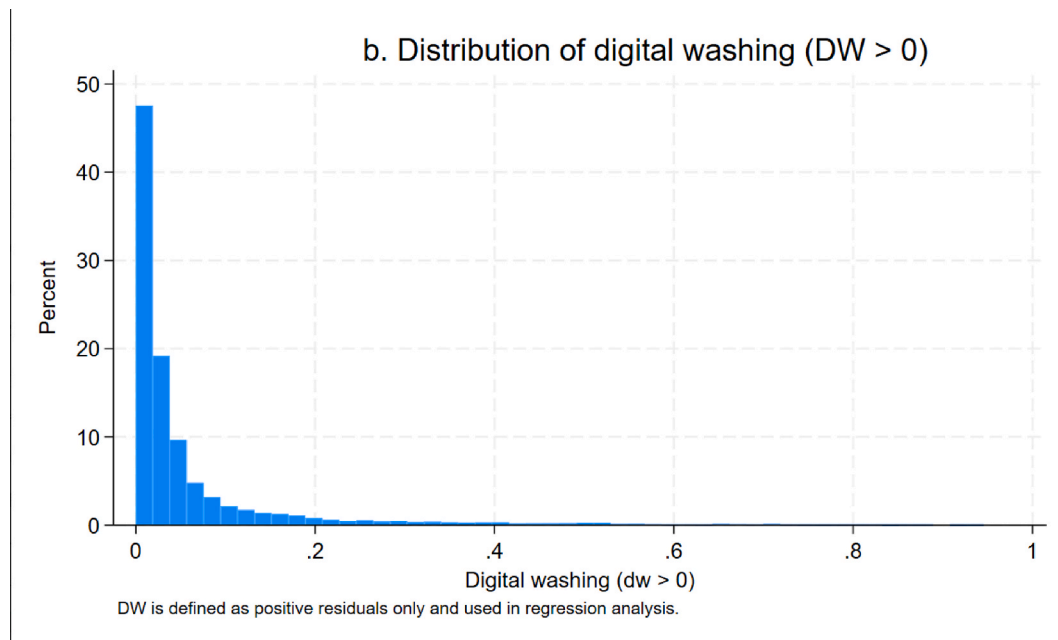


Fig. A1b. Distribution of digital washing (DW > 0)

Figure A2 further illustrates the industry-year dispersion of Digital Weight (DW) using boxplots for the top 15 industries by sample size. The results reveal significant heterogeneity in both the central tendency and dispersion of DW across industries and over time, confirming that, despite its concentrated overall distribution, DW retains substantial cross-sectional and temporal variation. Collectively, these figures demonstrate that DW captures economically and statistically informative differences in excessive digital disclosure intensity.

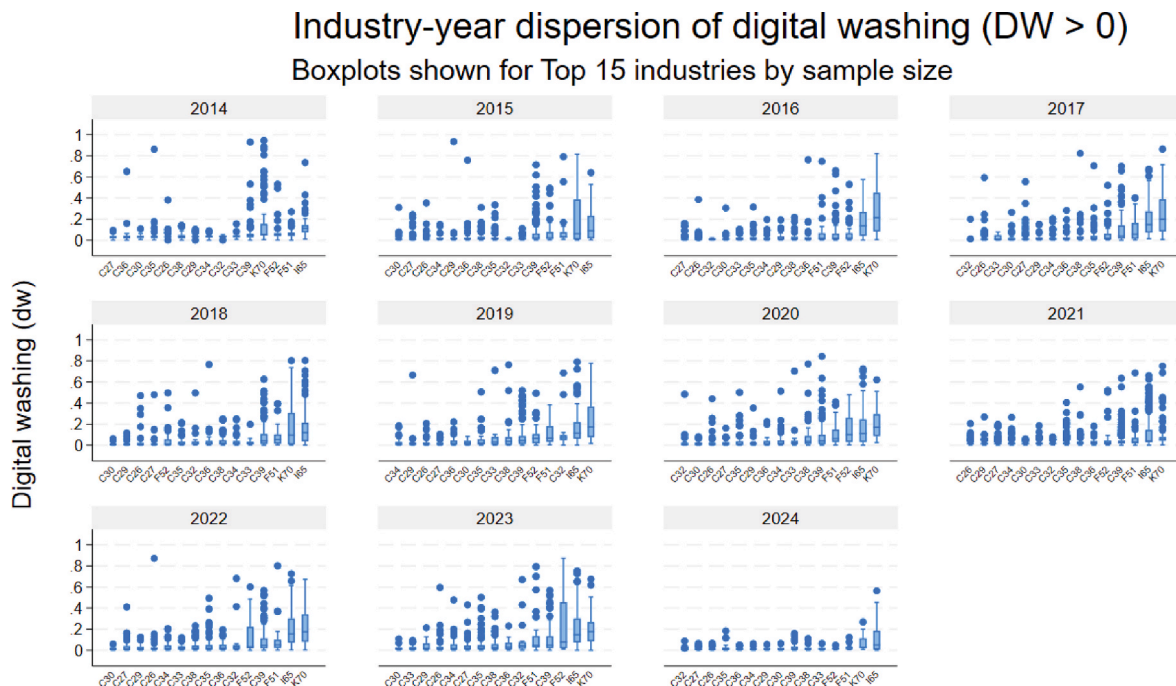


Fig. A2. Industry-year dispersion of digital washing (DW > 0)

Appendix B. Economic significance of digital washing on SG

Table 9
Economic significance of DW on SG

Change in digital washing	ΔSG	$\Delta SG / SD$
P25 $\rightarrow$ P75	-0.001512	-0.0245
P10 $\rightarrow$ P90	-0.004225	-0.0686

This table presents the economic magnitude of digital washing (DW) on corporate sustainable growth (SG). The values translate the estimated coefficients from the baseline regression into economically meaningful changes, utilizing the interquartile range (P25–P75) and a broader percentile contrast (P10–P90) of DW within the regression sample. The implied change in SG is denoted, and the effect is scaled by the sample standard deviation of SG ($SD = 0.0616$). All calculations are derived from the regression sample where $DW > 0$.

Data availability

Data will be made available on request.

References

Abideen, Z.U., Fuling, H., 2024. Sustainability reporting and investor sentiment. A sustainable development approach to Chinese-listed firms. *J. Clean. Prod.* 466, 142880. <https://doi.org/10.1016/j.jclepro.2024.142880>.

Akerlof, G.A., 1978. The market for “lemons”: quality uncertainty and the market mechanism. In: *Uncertainty in Economics*. Academic Press, pp. 235–251. <https://doi.org/10.1016/B978-0-12-214850-7.50022-X>.

Almon, S., 1965. The distributed lag between capital appropriations and expenditures. *Econometrica J. Econom. Soc.* 178–196. <https://doi.org/10.2307/1911894>.

Angrist, J.D., Krueger, A.B., 2001. Instrumental variables and the search for identification: from supply and demand to natural experiments. *J. Econ. Perspect.* 15 (4), 69–85. <https://www.aeaweb.org/articles?id=10.1257/jep.15.4.69>.

Bak-Coleman, J.B., Kennedy, I., Wack, M., Beers, A., Schafer, J.S., Spiro, E.S., Starbird, K., West, J.D., 2022. Combining interventions to reduce the spread of viral misinformation online. *Nat. Hum. Behav.* 6, 1372–1380. <https://doi.org/10.1038/s41562-022-01388-6>.

Barber, B.M., Odean, T., 2008. All that glitters: the effect of attention and news on the buying behavior of individual and institutional investors. *Rev. Financ. Stud.* 21 (2), 785–818. <https://doi.org/10.1093/rfs/hhm079>.

Benner, M.J., Tushman, M.L., 2003. Exploitation, exploration, and process management: the productivity dilemma revisited. *Acad. Manag. Rev.* 28 (2), 238–256. <https://doi.org/10.5465/amr.2003.9416096>.

Berg, Florian, Kölbel, Julian F., Rigobon, Roberto, 2022. Aggregate confusion: the divergence of ESG ratings. *Rev. Finance* 26 (6), 1315–1344. <https://doi.org/10.1093/rf/rfac033>.

Brynjolfsson, E., 2014. *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. 236. WW Norton Company.

Cao, Q., Zhou, Y., Du, H., Ren, M., Zhen, W., 2022. Carbon information disclosure quality, greenwashing behavior, and enterprise value. *Front. Psychol.* 13, 892415. <https://doi.org/10.3389/fpsyg.2022.892415>.

Chaudhuri, N., Roy, S., 2024. Misinformation on social platforms: a review and research agenda for governance. *Technol. Soc.* 76, 102460. <https://doi.org/10.1016/j.techsoc.2024.102460>.

Chen, X., Wan, P., Ma, Z., Yang, Y., 2024. Does corporate digital transformation restrain ESG decoupling? Evidence from China. *Hum. Soc. Sci. Commun.* 11 (1), 1–15. <https://www.nature.com/articles/s41599-024-02921-w>.

Connelly, B.L., Certo, S.T., Ireland, R.D., Reutzel, C.R., 2011. Signaling theory: a review and assessment. *J. Manag.* 37 (1), 39–67. <https://doi.org/10.1177/0149206310388419>.

Da, Z., Engelberg, J., Gao, P., 2011. In search of attention. *J. Finance* 66 (5), 1461–1499. <https://doi.org/10.1111/j.1540-6261.2011.01679.x>.

De la Boutetière, H., Montagner, A., Reich, A., 2018. *Unlocking Success in Digital Transformations*. 29. McKinsey & Company.

Filatotchev, I., Lanzolla, G., Syrigos, E., 2025. Impact of CEO's digital technology orientation and board characteristics on firm value: a signaling perspective. *J. Manag.* 51 (2), 875–912. <https://doi.org/10.1177/0149206323120081>.

Fioravante, R., 2024. Beyond the business case for responsible artificial intelligence: strategic CSR in light of digital washing and the moral human argument. *Sustainability* 16 (3), 1232. <https://doi.org/10.3390/su16031232>.

García-Sánchez, I.M., Hussain, N., Martínez-Ferrero, J., Ruiz-Barbadillo, E., 2019. Impact of disclosure and assurance quality of corporate sustainability reports on access to finance. *Corp. Soc. Responsib. Environ. Manag.* 26 (4), 832–848. <https://doi.org/10.1002/csr.1724>.

Grewal, J., Hauptmann, C., Serafeim, G., 2021. Material sustainability information and stock price informativeness. *J. Bus. Ethics* 171 (3), 513–544.

He, L., Ismail, K., 2024. ESG information disclosure, industrial policy support, debt financing costs. *Finance Res. Lett.* 66, 105630. <https://doi.org/10.1016/j.frl.2024.105630>.

Healy, P.M., Palepu, K.G., 2001. Information asymmetry, corporate disclosure, and the capital markets: a review of the empirical disclosure literature. *J. Account. Econ.* 31 (1–3), 405–440. [https://doi.org/10.1016/S0165-4101\(01\)00018-0](https://doi.org/10.1016/S0165-4101(01)00018-0).

Hsieh, C.-T., Klenow, P.J., 2009. Misallocation and manufacturing TFP in China and India. *Q. J. Econ.* 124 (4), 1403–1448. <https://doi.org/10.1162/qjec.2009.124.4.1403>.

Huszár, F., Ktena, S.I., O'Brien, C., Belli, L., Schlaikjer, A., Hardt, M., 2022. Algorithmic amplification of politics on Twitter. *Proc. Natl. Acad. Sci.* 119 (1), e2025334119. <https://doi.org/10.1073/pnas.2025334119>.

Ioannou, I., Serafeim, G., 2015. The impact of corporate social responsibility on investment recommendations: analysts' perceptions and shifting institutional logics. *Strateg. Manag. J.* 36 (7), 1053–1081. <https://doi.org/10.1002/smj.2268>.

Jamil, S.H., Khan, M.J., 2024. Do corporate environmental protection efforts reduce firm-level operating risk? Evidence from a developing country. *Bus. Strat. Environ.* 33 (5), 4480–4492. <https://doi.org/10.1002/bse.3711>.

Jensen, M.C., Meckling, W.H., 1976. Theory of the firm: managerial behavior, agency costs and ownership structure. *J. Financ. Econ.* 3 (4), 305–360. [https://doi.org/10.1016/0304-405X\(76\)90026-X](https://doi.org/10.1016/0304-405X(76)90026-X).

Ji, H., Sheng, S., Wan, J., 2024. Symbolic or substantive? The effects of the digital transformation process on environmental disclosure. *Systems* 12 (6), 197. <https://doi.org/10.3390/systems12060197>.

Ju, C., Fang, X., Shen, Z., 2024. Investor attention and corporate financialization: evidence from internet search volume. *Int. Rev. Financ. Anal.* 96, 103576. <https://doi.org/10.1016/j.irfa.2024.103576>.

Kahneman, D., 1973. *Attention and Effort*. 1063. Citeseer.

Kim, O., Verrecchia, R.E., 2001. The relation among disclosure, returns, and trading volume information. *Account. Rev.* 76 (4), 633–654. <https://doi.org/10.2308/accr-2001.76.4.633>.

Klein, L.R., 1955. Distributed lags and investment analysis. *Econ. J.* 65 (259), 523–525. <https://doi.org/10.2307/2227337>.

Kuo, Y.S., Huang, J.T., 2022. Factor-based investing in market cycles: Fama–French five-factor model of market interest rate and market sentiment. *J. Risk Finan. Manag.* 15 (10), 460. <https://doi.org/10.3390/jrfm15100460>.

Kuzma, E., Sehnem, S., 2023. Proposition of a structural model for business value creation based on circular business models, innovation, and resource recovery in the pet industry. *Bus. Strat. Environ.* 32 (1), 516–537. <https://doi.org/10.1002/bse.3158>.

Lamb, J.D., Tee, K.-H., 2024. Using stochastic frontier analysis instead of data envelopment analysis in modelling investment performance. *Ann. Oper. Res.* 332 (1), 891–907. <https://doi.org/10.1007/s10479-023-05428-w>.

Li, T., Shu, X., Liao, G., 2024. Does corporate greenwashing affect investors' decisions? *Fin. Res. Lett.* 67, 105877. <https://doi.org/10.1016/j.frl.2024.105877>.

Lin, X., She, G., Yoon, A., Zhu, H., 2025. Shareholder value implications of supply chain ESG: evidence from negative incidents. *Rev. Account. Stud.* 1–33. <https://link.springer.com/article/10.1007/s11142-025-09903-6>.

Liu, C., Zhao, X., 2025. Earnings pressure and corporate social responsibility impression management. *Serv. Sci.* 17 (1), 18–34. <https://doi.org/10.1287/serv.2023.0052>.

Liu, F., Li, L., Ye, B., Qin, Q., 2023. A novel stochastic semi-parametric frontier-based three-stage DEA window model to evaluate China's industrial green economic efficiency. *Energy Econ.* 119, 106566. <https://doi.org/10.1016/j.eneco.2023.106566>.

Liu, Z., Zhou, J., Li, J., 2023. How do family firms respond strategically to the digital transformation trend: disclosing symbolic cues or making substantive changes? *J. Bus. Res.* 155, 113395. <https://doi.org/10.1016/j.jbusres.2022.113395>.

Liu, Y., Heinberg, M., Huang, X., Eisingerich, A.B., 2023. Building a competitive advantage based on transparency: when and why does transparency matter for corporate social responsibility? *Bus. Horiz.* 66 (4), 517–527. <https://doi.org/10.1016/j.bushor.2022.10.004>.

Long, L., Wang, C., Zhang, M., 2025. Does social media pressure induce corporate hypocrisy? Evidence of ESG greenwashing from China. *J. Bus. Ethics* 197 (2), 311–338. <https://link.springer.com/article/10.1007/s10551-024-05747-3>.

Luo, S., Liu, J., 2024. Enterprise service-oriented transformation and sustainable development driven by digital technology. *Sci. Rep.* 14 (1), 10047. <https://www.nature.com/articles/s41598-024-60922-w>.

- McLoughlin, K.L., et al., 2024. Human–algorithm interactions help explain the spread of misinformation on social media. *Patterns* 5 (7), 100942. <https://doi.org/10.1016/j.patter.2024.100942>.
- Meyer, J.W., Rowan, B., 1977. Institutionalized organizations: formal structure as myth and ceremony. *Am. J. Sociol.* 83 (2), 340–363. <https://doi.org/10.1086/226550>.
- Neokleous, C.I., Elmarzouky, M., 2025. Strategic sustainability reporting, impression management and board gender diversity: evidence from luxury fashion industry. *Bus. Strat. Environ.* <https://doi.org/10.1002/bse.70253>.
- Nygaard, A., Silkoet, R., 2023. Sustainable development and greenwashing: how blockchain technology information can empower green consumers. *Bus. Strat. Environ.* 32 (6), 3801–3813. <https://doi.org/10.1002/bse.3338>.
- Ocasio, W., 1997. Towards an attention-based view of the firm. *Strateg. Manag. J.* 18 (S1), 187–206. [https://doi.org/10.1002/\(SICI\)1097-0266\(199707\)18:1+<187::AID-SMJ936>3.0.CO;2-K](https://doi.org/10.1002/(SICI)1097-0266(199707)18:1+<187::AID-SMJ936>3.0.CO;2-K).
- Oludapo, S., Carroll, N., Helfert, M., 2024. Why do so many digital transformations fail? A bibliometric analysis and future research agenda. *J. Bus. Res.* 174, 114528. <https://doi.org/10.1016/j.jbusres.2024.114528>.
- Omonijo, O.N., Zhang, Y., 2025. Examining the relationship between technological innovation, environmental social governance and corporate sustainability: the moderating role of green operational innovation. *Humanit. Soc. Sci. Commun.* 12, 107. <https://doi.org/10.1057/s41599-025-04389-8>.
- Orlikowski, W.J., Barley, S.R., 2001. Technology and institutions: what can research on information technology and research on organizations learn from each other? *MIS Q.* 145–165. <https://doi.org/10.2307/3250927>.
- Papa, M., Wiczorek-Kosmala, M., Losa, A., Swalek, A., 2024. The impact of ESG regulation on environmental decoupling—an exploratory study on Polish listed companies. *Sustainability* 16 (17), 7309. <https://doi.org/10.3390/su16177309>.
- Parmeter, C.F., Kumbhakar, S.C., 2025. The generalized panel data stochastic frontier model: a review and nonparametric estimation. *J. Prod. Anal.* 1–19. <https://doi.org/10.1007/s11123-025-00769-z>.
- Quttainah, M.A., Ayadi, I., 2024. The impact of digital integration on corporate sustainability: emissions reduction, environmental innovation, and resource efficiency in the European. *J. Innov. Knowl.* 9 (3), 100525. <https://doi.org/10.1016/j.jik.2024.100525>.
- Richardson, S., 2006. Over-investment of free cash flow. *Rev. Account. Stud.* 11 (2), 159–189. <https://doi.org/10.1007/s11142-006-9012-1>.
- Song, P., Gu, Y., Su, B., Tanveer, A., Peng, Q., Gao, W., et al., 2023. The impact of green technology research and development (R&D) investment on performance: a case study of listed energy companies in Beijing, China. *Sustainability* 15 (16), 12370. <https://doi.org/10.3390/su151612370>.
- Spence, M., 1973. Job market signaling. *Q. J. Econ.* 87 (3), 355–374. <https://doi.org/10.2307/1882010>.
- Stock, J.H., Yogo, M., 2002. Testing for weak instruments in linear IV regression. <http://www.nber.org/papers/t0284>.
- Strange, N., Bogetoft, P., Aalmo, G.O., Talbot, B., Holt, A.H., Astrup, R., 2021. Applications of DEA and SFA in benchmarking studies in forestry: state-of-the-art and future directions. *Int. J. For. Eng.* 32 (Suppl. 1), 87–96. <https://doi.org/10.1080/14942119.2021.1914809>.
- Suchman, M.C., 1995a. Managing legitimacy: strategic and institutional approaches. *Acad. Manag. Rev.* 20 (3), 571–610. <https://doi.org/10.2307/258788>.
- Suchman, M.C., 1995b. Managing legitimacy: strategic and institutional approaches. *Acad. Manag. Rev.* 20 (3), 571–610. <https://journals.aom.org/doi/abs/10.5465/AMR.1995.9508080331>.
- Tabrizi, B., Lam, E., Girard, K., Irvin, V., 2019. Digital transformation is not about technology. *Harv. Bus. Rev.* 13 (March), 1–6. [https://doi.org/10.1016/0737-6782\(96\)83110-9](https://doi.org/10.1016/0737-6782(96)83110-9).
- Tolentino-Zondervan, F., DiVito, L., 2024. Sustainability performance of Dutch firms and the role of digitalization: the case of textile and apparel industry. *J. Clean. Prod.* 459, 142573. <https://doi.org/10.1016/j.jclepro.2024.142573>.
- Tone, K., Tsutsui, M., 2009. Network DEA: a slacks-based measure approach. *Eur. J. Oper. Res.* 197 (1), 243–252. <https://doi.org/10.1016/j.ejor.2008.05.027>.
- Velte, P., 2023. Determinants and consequences of corporate social responsibility decoupling—Status quo and limitations of recent empirical quantitative research. *Corp. Soc. Responsib. Environ. Manag.* 30 (6), 2695–2717. <https://doi.org/10.1002/csr.2538>.
- Venkatraman, N., Camillus, J.C., 1984. Exploring the concept of “fit” in strategic management. *Acad. Manag. Rev.* 9 (3), 513–525. <https://doi.org/10.5465/amr.1984.4279696>.
- Vial, G., 2021. Understanding digital transformation: a review and a research agenda. *Manag. Digital Transform.* 13–66.
- Wang, H.-J., Ho, C.-W., 2010. Estimating fixed-effect panel stochastic frontier models by model transformation. *J. Econom. J. Econom.* 157 (2), 286–296. <https://doi.org/10.1016/j.jeconom.2009.12.006>.
- Wei, C., Zhou, H., Zhang, X., Ouyang, S., 2025. The dark side of digital transformation: digital washing and accounting conservatism. *Int. Rev. Econ. Finance* 98, 103933. <https://doi.org/10.1016/j.iref.2025.103933>.
- Weritz, P., Braojos, J., Matute, J., Benitez, J., 2025. Impact of strategic capabilities on digital transformation success and firm performance: theory and empirical evidence. *Eur. J. Inf. Syst.* 34 (3), 415–435. <https://doi.org/10.1080/0960085x.2024.2311137>.
- Westphal, J.D., Zajac, E.J., 1998. The symbolic management of stockholders: corporate governance reforms and shareholder reactions. *Adm. Sci. Q.* 127–153. <https://doi.org/10.2307/2393593>.
- Wooldridge, J.M., 2002. *Econometric analysis of cross section and panel data*. MIT press, Cambridge, MA 108 (2), 245–254.
- Xu, X.L., Li, J., Wu, D., Zhang, X., 2021. The intellectual capital efficiency and corporate sustainable growth nexus: comparison from agriculture, tourism and renewable energy sector. *Environ. Dev. Sustain.* 23 (11), 16038–16056. <https://doi.org/10.1007/s10668-021-01319-x>.
- Yezegel, A., 2023. The value of eliciting information: evidence from sell-side analysts. *Account. Rev.* 98 (3), 459–486. <https://doi.org/10.2308/tar-2020-0308>.
- Yoshikuni, A.C., Dwivedi, R., Kamal, M.M., Zhou, D., Dwivedi, P., Apolinario, S., 2024. A dynamic information technology capability model for fostering innovation in digital transformation. *J. Innov. Knowl.* 9 (4), 100589. <https://doi.org/10.1016/j.jik.2024.100589>.
- Yu, H., Zhang, J., 2025. Do green investors empower companies to develop sustainably? A study based on the perspective of innovation investment and corporate governance levels. *Finance Res. Lett.* 79, 107263. <https://doi.org/10.1016/j.frl.2025.107263>.
- Yuan, W., Zhang, M., Guo, X., Jiang, Y., 2025. ESG disclosure and firm long-term value: the mediating effect of tax avoidance. *Finance Res. Lett.*, 107728 <https://doi.org/10.1016/j.frl.2025.107728>.
- Zhang, C., Chen, P., Hao, Y., 2022. The impact of digital transformation on corporate sustainability-new evidence from Chinese listed companies. *Front. Environ. Sci.* 10, 1047418. <https://doi.org/10.3389/fenvs.2022.1047418>.
- Zhang, K., Pan, Z., Zhang, K., Ji, F., 2023. The effect of digitalization transformation on greenwashing of Chinese listed companies: an analysis from the dual perspectives of resource-based view and legitimacy. *Front. Environ. Sci.* 11, 1179419. <https://doi.org/10.3389/fenvs.2023.1179419>.
- Zhou, G., Liu, L., Luo, S., 2022. Sustainable development, ESG performance and company market value: mediating effect of financial performance. *Bus. Strat. Environ.* 31 (7), 3371–3387. <https://doi.org/10.1002/bse.3089>.